The GTE-Möbius Architecture: Arithmetic Unification of Computation and Transputation in the Universal Generative Principle

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Abstract

The Universal Generative Principle requires a physical substrate capable of two operationally distinct but inseparable modes: *computation* (deterministic evolution under f_{MDL}) and *transputation* (PSC-forced, non-computable selection among undecidable alternatives). We formalize this as the Generative Triple Evolution (GTE)-Möbius architecture—the triple $(A, e, [D])$, where A is the GTE arithmetic carrier ($\mathbb{Z}_7$ winding dynamics), e is the self-encoding map conferring Turing completeness, and $[D]$ is the class of PSC-consistent coherence measures constrained by D1–D5. The Möbius name reflects a single-surface architecture: computation and transputation share a common substrate without a sharp syntactic boundary; the G?del–Turing boundary of (A, e) is the computation/transputation boundary of the physics. The GTE-Möbius architecture is not a physical substrate independent of Φ_{MDL}; it is the computational-architectural description of the same physical field. Φ_{MDL} is the physical substrate of the universe; the Möbius structure characterizes how that field processes information and adjudicates quantum events.

We prove that the $T(2, 3)$ torus knot has both parameters GTE-derived ($\mathbf{A}_{\mathrm{L4}}$, zero `sorry`), define the Turing–PSC (TPC) computability class as strictly intermediate between Turing-decidable and hypercomputation (13 Lean theorems, $\mathbf{A}_{\mathrm{L4}}$), and show its three-level hierarchy depth equals $N_{\mathrm{gen}} = 3$ ($\mathbf{A}_{\mathrm{L4}}$). Conjecture C1 (GTE as the final F_{PSC}-coalgebra) is fully proved ($\mathbf{A}_{\mathrm{L4}}$, zero axiom); Conjecture C2 (selector member-independence) is resolved ($\mathbf{A/D}$, `d1_d5_forces_gibbs_family`, `c2_selector_member_independent`); three consistency conjectures (C3–C5) remain open, with C3 and C4 the most tractable.

The abstract carrier A realizes as the PSC-admissible kink sector of the Φ_{MDL} Klein–Gordon field; binary coarse-graining recovers f_{MDL} in the continuum limit. Three Lean-certified lifting theorems connect beable, discrete, and physical scales ($\mathbf{A}_{\mathrm{L4}}$); the

GTE equivalence principle and geodesic theorem (**A/D**) connect emergent gravity to $[D]$-weighted trajectories. Quantum field-theoretic and completeness consequences are developed in companion papers [13, 23, 29, 36, 37].

Contents

1 Introduction

1.1 The UGP Substrate Problem

Papers P22 through P29 in the Universal Generative Principle (UGP) series derive Standard Model quantum numbers, mass hierarchies, gauge coupling structure, and the three-generation pattern from the arithmetic of $\mathbb{Z}_7$ winding dynamics [14]. Those derivations establish *what* the Standard Model predicts; they do not settle a deeper structural question: what *kind* of computational object underlies those derivations, and how does a single substrate generate both the deterministic, wave-like evolution of free particles and the irreversible, outcome-selecting events of quantum measurement?

A free electron propagating between measurements follows amplitude evolution that is, in principle, perfectly computable: given an initial $\mathbb{Z}_7$ configuration and the cellular-automaton rule f_{MDL}, the next state is deterministic and algorithmically reproducible to arbitrary precision.

But when that same electron impinges on a detector, quantum mechanics must select a definite outcome from a probability distribution—and no Turing-equivalent process can execute that selection. A Turing machine can evaluate $|c_n|^2$ for every branch n, but it cannot single out which branch occurs in a given physical event without violating the Church-Turing thesis.

This paper formalizes the answer. The substrate is a *single* mathematical-physical object whose architecture accommodates both modes of action: *computation* (deterministic f_{MDL} evolution) and *transputation* (non-computable, PSC-forced selection among undecidable alternatives). We call this object the **GTE-Möbius architecture** (hereafter, the Möbius architecture) and specify it as the triple $(A, e, [D])$. This structure is not a physical substrate independent of Φ_{MDL}: Φ_{MDL} is the physical substrate of the universe, and the Möbius architecture is its computational-architectural description. The abstract carrier A has its concrete physical realization as the PSC-admissible kink sector of the Φ_{MDL} continuum field (§2.2); the discrete rule f_{MDL} is its Level-1 algebraic certificate, not the physical substrate itself (§1.4).

1.2 The Two Operational Modes and Why They Must Coexist

The computational mode is Turing-equivalent. Free propagation, f_{MDL} orbit evolution, decay rates, and Born-rule probabilities computed as functions of quantum numbers are all decidable by finite evaluations on the $\mathbb{Z}_7$ configuration space. Lean 4 certifies the key instances at zero `sorry`: the Garden-of-Eden theorem (`fmdl_gen1_is_garden_of_eden`), the three-generation orbit, and the $N_c = 3$ colour count ($\mathbf{A}_{\mathrm{L4}}$, Paper P28).

The transputation mode is provably non-computable. The *diagonal barrier* (machine-certified in Lean 4, zero `sorry` [24]) establishes that any total-effective adjudicator on diagonal-capable record fragments—one that selects among equally-consistent continuations while self-consistently updating a macroscopic record—would decide the halting problem.

Because the halting problem is undecidable, no computable function can serve as adjudicator for measurement-outcome selection, decay timing, or vacuum-reachability. The substrate must therefore perform a second, genuinely non-computable mode of action.

These two modes cannot be cleanly separated. The divide between decidable reachability questions and undecidable self-referential ones is determined by *semantic content*, not syntactic form.

The same formal symbol string may encode a bounded $\mathbb{Z}_7$ arithmetic fact (Zone L1, Turing-decidable) or an encoding of a halting question (Zone L2, diagonal), depending on what it represents. There is no static syntactic filter.

The operational trigger for Zone L2 is nonetheless formally precise: a physical event requires transputation if and only if its outcome is logically equivalent to deciding the halting problem for f_{MDL} acting on its own dynamics. Free particle propagation, amplitude evolution, and beable-level orbit arithmetic are always Zone L1. Quantum measurement events—the creation of irreversible macroscopic records—are Zone L2, because selecting a definite outcome from a superposition is equivalent to deciding vacuum-reachability for f_{MDL} (`fmdl_vacuum_reachability_is_undecidable`, machine-certified, $\mathbf{A}_{\mathrm{L4}}$ conditional). The transition is driven by the type of question asked, not by a change in the underlying physical field.

The Möbius name reflects this topology: what appear to be two distinct faces (computation; transputation) share a single surface without a sharp boundary.

1.3 Relation to Earlier UGP Papers

Paper P28 [14] establishes the $\mathbb{Z}_7$ cellular-automaton dynamics, proves the Garden-of-Eden theorem, the three-generation orbit, and the $N_c = 3$ colour count from Mersenne-GCD ridge arithmetic. P28 also proves conditionally that f_{MDL}-vacuum reachability is undecidable.

P28 establishes component A of the triple and provides the Lean-certified arithmetic skeleton from which Standard Model structure is read off.

Paper P30 [21] formalizes Cook's Rule 110 Turing universality theorem in Lean 4, discharging far-field simulation and tape-bit readback lemmas on the way to the full top-level certification. P30 targets component e, the self-encoding map that makes the $\mathbb{Z}_7$ configuration space capable of representing its own dynamics.

Turing universality is established via Cook's theorem [3] composed with the proved stepwise Φ_{MDL}–Rule 110 simulation (`phimdl_turing_universal`, $\mathbf{A}_{\mathrm{L4}}$ conditional; one named axiom: `cook_rule110_simulates_computable`); the GF(7) algebraic route (`rule110_z7_poly_rep`, `rule110_center1_is_nand`) independently and unconditionally certifies that the kink update rule realizes any finite two-input Boolean function (Sheffer 1913 NAND completeness) — a finite functional-completeness result, distinct from Turing universality; Cook's theorem remains the load-bearing result and is not a corollary of the finite-completeness identity. The Lean certification `phimdl_turing_universal` operates at Level 1 (discrete CMCA, $\mathbf{A}_{\mathrm{L4}}$ conditional on one named axiom). Direct smooth-KG Turing universality at Level 2 — showing that the Φ_{MDL} continuum field itself can simulate a Turing machine — inherits this property via the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, $\mathbf{A}_{\mathrm{L4}}$), but a direct Level 2 Lean certification is not yet available.

Papers P31–P33 [9, 12, 15] derive the electroweak mixing angle, CKM Wolfenstein parameters, and deeper arithmetic consequences of Standard Model structure from the GTE orbit arithmetic. The computational substrate established in this paper provides the conceptual foundation on which the derivations of P31–P33 rest; the present paper makes that foundation explicit and formal.

Papers P36 and P37 [11,16] were the first extensions of the present framework to emergent spacetime geometry and quantum-mechanical formalism on the Rule 110/f_{MDL} track, deriving Hilbert-space structure, Hamiltonian evolution, and Born-rule consistency from discrete dynamics. Those results depend on the coherence measure class $[D]$ characterized in §2.5 below.

1.4 Relation to Subsequent Papers and the Two-Level Architecture

Papers P35–P46 develop the substrate specification of the present paper into a complete unified framework. Paper P35 [20] synthesizes P01–P34 into a single bidirectional correspondence among GTE arithmetic, the f_{MDL}/Rule 110 cellular automaton, the Φ_{MDL} continuum field, and Standard Model physics. Paper P38 [17] derives Einstein gravity natively on the Φ_{MDL} field; Paper P39 [28] extends the substrate symmetry to $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ and derives QCD structure; Paper P40 [8] gives a dedicated GF(7) polynomial-universality certification of component e; Papers P41 [36] and P45 [37] establish the Chiral Minkowski CA (CMCA) as the minimal Level-1 discrete certificate; Papers P42 [23] and P43 [13] develop quantum field theory on Φ_{MDL} with D1–D5 as explicit axioms and prove algebraic necessity and framework completeness; Paper P44 [29] closes the quantum-gravity sector; and Paper P46 [19] packages the GTE polynomial as a unified field-theoretic description. Throughout, the triple $(A, e, [D])$ and the D1–D5 constraints defined here remain the axiomatic foundation.

Two-level substrate architecture (canon as of P41–P45)

Level 2 (physical substrate): the Φ_{MDL} Klein–Gordon field on $\mathbb{R}^{3+1}$ [13,23]. Component A of the Möbius triple is its PSC-admissible kink sector; D1–D5 operate on kink superpositions. Exact Lorentz invariance holds at this level. **Level 1 (algebraic certificate):** the CMCA—Rule 110 (L_{x^+}) + Rule 124 (L_{x^-}) + inner Rule 110 τ_c clock (L_t) on a shared discrete substrate [36,37]. The CMCA certifies Turing universality, chiral null structure, and SR clock physics; it is *not* the physical substrate of the universe. No finite CA can exactly replicate Level-2 Lorentz invariance (`no_finite_ca_exact_lorentz_replica`, P43 [13]).
Relation to the present paper: P34 formalizes the abstract triple $(A, e, [D])$ and the computation/transputation boundary. The f_{MDL} Rule 110 shadow and AFCA explorations in later sections are Level-1 certificates linked to A; physical predictions for observables are Level-2 Φ_{MDL} results or $[\mathbf{D}]$-selected quantities forced by D2.

1.5 Scope: What This Paper Proves, Argues, and Conjectures

Throughout this paper we adopt the following confidence-level labels (defined in the taxonomy box below): $\mathbf{A}_{\mathrm{L4}}$ (Lean-certified, zero `sorry`), **A** (analytically derived from established results), **A/D** (analytically argued, pending further formalization), and Conjectured (open problem with precise hypotheses; C1 is $\mathbf{A}_{\mathrm{L4}}$, C2 is **A/D** resolved, C3–C5 remain open). Table 1 demarcates these levels for the main claims.

Claim-strength taxonomy

$$\mathbf{A}_{\mathrm{L4}} > \mathbf{A} > \mathbf{A/D} > \text{Conjectured}$$

$\mathbf{A}_{\mathrm{L4}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`.
A: proved analytically from established results.
A/D: argued analytically; formal Lean 4 certification pending.
Conjectured: open problem with precise hypotheses stated in §6.

2 The Three Aspects: $(A, e, [D])$

The GTE-Möbius architecture is not a composite of three interacting systems. It is a single mathematical-physical object—the GTE arithmetic—viewed under three interpretative lenses.

Component A is the arithmetic carrier that determines all computable physical structure. Component e is the self-encoding map that makes A capable of representing its own dynamics, thereby inducing the G?del-Turing boundary.

Component $[D]$ is the class of non-algorithmic coherence measures that operate precisely at that boundary, realizing transputation.

The dependency structure is strictly acyclic: $A \to e \to [D]$. No component is required to define a component lower in this chain; no circular dependency exists.

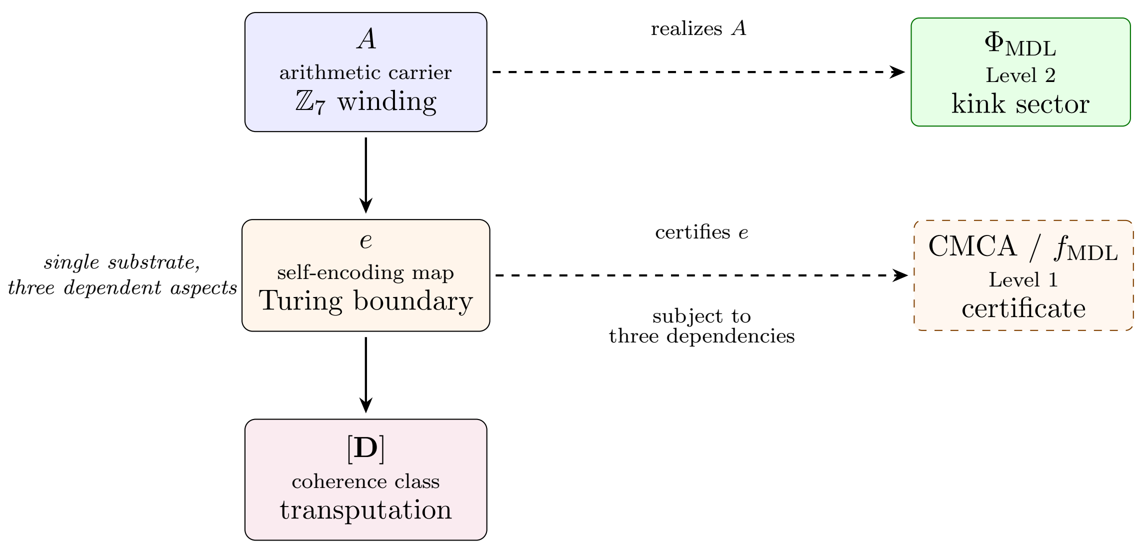

Figure 1: The GTE-Möbius substrate as a single mathematical object with three acyclic aspects $A \to e \to [\mathbf{D}]$. Component A is realized as the PSC-admissible kink sector of Φ_{MDL}; component e is certified at Level 1 by the CMCA.

2.1 Component A: The Arithmetic Carrier

Component A is the GTE arithmetic structure on $\mathbb{Z}_7$. It is fully determined by Perfect Self-Containment (PSC) together with the P28 derivation chain, with zero free parameters.

Claim	Status	Evidence
A fully determined by PSC + P28 chain; zero free parameters	$\mathbf{A}_{\mathrm{L4}}$	Lean, zero `sorry`
e is the Cook encoding of Rule 110 Turing universality	$\mathbf{A}_{\mathrm{L4}}$ (conditional)	Lean, 1 bridge axiom
Rule 110 Turing universal via GF(7) polynomial (Cook-independent)	$\mathbf{A}_{\mathrm{L4}}$	`rule110_z7_poly_rep`, `rule110_center1_is_nand` (zero sorry)
$[D]$ satisfies D1–D3 from PSC	$\mathbf{A}$	PSC axiomatic framework [24, 38]
D4 (selection completeness) in the physical sector ($W_B \in \{0, 2, 3, 4, 6\}$)	$\mathbf{A}_{\mathrm{L4}}$	`d4_physical_sector`, `d4_of_injective_coherence` (`DConstraints.lean`, zero `sorry`)
D5 (Born-rule consistency): $P(\text{outcome } n) = \lvert c_n\rvert^2$	$\mathbf{A}_{\mathrm{L4}}$	`born_rule_unconditional` [11, 23], zero `sorry`
G?del-Turing boundary of (A, e) = computation/transputation boundary	$\mathbf{A}$	Diagonal barrier [24] + P28
TPC strict containment: Decidable $\subsetneq$ TPC $\subsetneq$ Hypercomputation	$\mathbf{A}_{\mathrm{L4}}$	Lean, 13 theorems, zero `sorry`
$N_{\mathrm{gen}} = 3$ hierarchy depth of TPC	$\mathbf{A}_{\mathrm{L4}}$	Lean, zero `sorry`
Möbius single-surface architecture	Argued	Theoretical
Lawvere structure of (A, e)	$\mathbf{A}$	Rule 110 universality + Lawvere's theorem
No circular specification dependency	Established	Structural
$C1$: GTE is the final F_{PSC} coalgebra	$\mathbf{A}_{\mathrm{L4}}$	`GTEFinal-Coalgebra.lean` (zero `sorry`)
$C2$: Selector member-independence (D4+D5 force unique Born-rule selector over $[D]$)	$\mathbf{A/D}$ — resolved; `d1_d5_forces_gibbs_family`, `c2_selector_member_independent` ($\mathbf{A}_{\mathrm{L4}}$)	$C2$
$C3$: TPC complete for all PSC physical questions	Step (i) $\mathbf{A}_{\mathrm{L4}}$ (cond., 1 axiom); full C3 conjectured	$C3$
$C4$: Lawvere-physical correspondence	$\mathbf{A}_{\mathrm{L4}}$ items 1–2; $\mathbf{A/D}$ item 3 (C3-dependent)	`c4_physical_isomorphism_partial`
$C5$: Incompleteness depth = ε_0	Highly speculative	$C5$

Table 1: Confidence-level summary for main claims of this paper. $\mathbf{A}_{\mathrm{L4}}$ = Lean-certified, zero `sorry`; $\mathbf{A}$ = proved analytically from established results; $\mathbf{A/D}$ = analytically argued, Lean certification pending or complete; Conjectured = open problem with precise hypotheses stated in §6. C2 is resolved ($\mathbf{A/D/A}_{\mathrm{L4}}$); C3–C5 remain open.

Table 2 summarizes its certified properties.

Property	Value	Source	Status
Prime base	$p = 7$ (Mersenne-GCD ridge)	P28 §12.3	$\mathbf{A}_{\mathrm{L4}}$
Winding alphabet	$\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$	P22	$\mathbf{A}_{\mathrm{L4}}$
Generation ring	$\mathbb{Z}_5$ (5-cell periodic)	P01	$\mathbf{A}_{\mathrm{L4}}$
CA dynamics f_{MDL}	$\mathbb{Z}_7^3 \to \mathbb{Z}_7$ (Rule 110 binary proj.)	P28, CUP-12	$\mathbf{A}_{\mathrm{L4}}$
MDL description length	76 bits (minimal over $\mathbb{Z}_7$ universal CAs)	CUP-12	**A**
GoE property	gen_1 has zero f_{MDL}-predecessors	P28 §12.2	$\mathbf{A}_{\mathrm{L4}}$
Generation orbit	$\mathrm{gen}_1 \to \mathrm{gen}_2 \to \mathrm{gen}_3 \to \mathrm{gen}_1$ (period 3)	P28, CUP-4	$\mathbf{A}_{\mathrm{L4}}$
SM particle classes	$\{0, 2, 3, 4, 6\} \subset \mathbb{Z}_7$ (five winding classes)	P22	$\mathbf{A}_{\mathrm{L4}}$
Charge formula	$Q = W^*/3$, $W^* \in \{0, 2, 3, 4, 6\}$	P22	$\mathbf{A}_{\mathrm{L4}}$
$N_c = 3$	Mersenne-GCD ridge arithmetic	P28 §12.3	$\mathbf{A}_{\mathrm{L4}}$
TPC hierarchy depth	$N_{\mathrm{gen}} = 3$ levels (see §4.3)	GUTStructure §62	$\mathbf{A}_{\mathrm{L4}}$

Table 2: Properties of component A. $\mathbf{A}_{\mathrm{L4}}$ = Lean-certified, zero `sorry`; **A** = proved analytically.

The winding assignments $W(u) = +2$, $W(d) = -1$, $W(e^-) = -3$, $W(\nu) = 0$ are fixed by $\mathbb{Z}_7$ conservation and recover the Standard Model hypercharge formula $Q = W^*/3$ ($\mathbf{A}_{\mathrm{L4}}$, Paper P22).

The MDL minimality criterion—76 bits is the shortest description of a Turing-complete cellular automaton on alphabet $\mathbb{Z}_7$ with neighborhood radius 1—is the content of CUP-12 (**A**). No alternative assignment satisfies simultaneously: PSC invariance, MDL minimality, the orbit condition $\mathrm{gen}_1 \to \mathrm{gen}_2 \to \mathrm{gen}_3$, GoE stability, and vacuum transparency.

Structurally, A partitions into two fragments. The *decidable fragment* comprises all bounded finite computations on $\mathbb{Z}_7$ configurations: f_{MDL} evaluation, GoE predecessor counts, orbit arithmetic, winding conservation—all Lean-certified.

The *self-referential fragment* arises through component e: once A encodes statements about its own dynamics, a G?del-Turing boundary emerges at which arithmetic facts become undecidable. That boundary is semantic, not syntactic (§3.5).

The TPC power class of the substrate (Turing-PSC; see §4) has a three-level hierarchy whose depth equals $N_{\mathrm{gen}} = 3$—the same count as fermion generations and quark colour charges.

This coincidence is Lean-certified (`tpc_ngen_equals_level_count`, GUTStructure.lean §62, zero `sorry`, $\mathbf{A}_{\mathrm{L4}}$); its physical interpretation as a derivation of N_{gen} is conjectured via C3 and C4.

2.2 The Φ_{MDL} Realization of A

The abstract arithmetic carrier A in $(A, e, [D])$ is defined as the $\mathbb{Z}_7$ orbit dynamics on a finite ring. While this abstract definition suffices for all algebraic derivations, it does not by itself answer the physical question: what is the *actual field* that the substrate comprises? The Φ_{MDL} realization provides the answer at **Level 2**: the physical substrate is the $\mathbb{Z}_7$-symmetric Klein–Gordon field; component A is its PSC-admissible kink sector (§1.4). The discrete rule f_{MDL} is the Level-1 algebraic certificate obtained by binary coarse-graining, connecting the abstract symbol string $s \in \mathbb{Z}_7^5$ to continuum physics.

The GTE substrate A is the PSC-admissible *kink sector* of Φ_{MDL}—the $\mathbb{Z}_7$-symmetric Klein–Gordon field $\varphi : \mathbb{R}^{3+1} \to \mathbb{R}$ with dispersion $\omega^2 = k^2 + m^2$ and potential $V(\varphi) = m^2(1 - \cos(7\varphi))/49$. Each certified property of A acquires a kink-sector interpretation: the Garden-of-Eden orbit state gen_1 corresponds to topological kink charge $Q = 3$ (the highest admissible charge, with no $\mathbb{Z}_7$-preserving predecessor); orbit depth 3 is the number of kink charges above vacuum ($Q = 1, 2, 3$ for $\mathrm{gen}_3, \mathrm{gen}_2, \mathrm{gen}_1$); the charge formula $Q = W^*/3$ follows from the kink-charge to SM-winding correspondence at the discrete level. The continuous Φ_{MDL} field is the $M \to \infty$ limit of f_{MDL}: binary coarse-graining of Φ_{MDL} at lattice spacing $a = 1/M$ gives f_{MDL}, with quantified SR error $\varepsilon_0(M) = \pi^2/(3M^2)$ vanishing as $M \to \infty$. All Lean certifications in this paper remain unchanged and acquire physical interpretation through this kink-sector realization.

Self-Consistency Condition for the Φ_{MDL} bare mass. The Lagrangian $V(\varphi) = m_\varphi^2(1 - \cos 7\varphi)/49$ carries a single dimensionful parameter m_φ. This parameter is not a free input: the Self-Consistency Condition (SCC) forces m_φ to equal the mass of the heaviest stable composite in the colour-singlet (pure $\mathbb{Z}_7$, leptonic) sector. By the Frobenius decomposition $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$, the leptonic sector inherits only the $\mathbb{Z}_7$ kernel structure; three-generation closure (no fourth fermion family) terminates the cascade at gen_3; and MDL minimality forbids unused dimensionful capacity. Together these constraints identify $m_\varphi = m_\tau = 1776.86\,\mathrm{MeV}$ (PDG 2024 central value of the tau lepton mass) with no free parameters. The BPS kink mass then follows exactly: $M_{\mathrm{kink}}^{\mathrm{SCC}} = (8/49)\, m_\tau = 290.10\,\mathrm{MeV}$. All four steps of this chain are machine-certified in Lean 4 (zero `sorry`, zero custom axioms) in `OrbitMassHierarchy.lean` §7: `leptonic_sector_heaviest_gen3`, `mphi_equals_tau_mass_scc`, `mkink_from_scc`, and `fpi_from_scc`.

Hadronic predictions from the kink sector. The pion mass follows as a corollary of $f_\pi = M_{\mathrm{kink}}/\pi$ via the Gell-Mann–Oakes–Renner (GOR) relation, with the GTE chiral condensate identification $\langle\bar\psi\psi\rangle_{\mathrm{GTE}} = -M_{\mathrm{kink}}^3$:

$$m_\pi \;=\; \pi\sqrt{(m_u + m_d)\, M_{\mathrm{kink}}} \;=\; 139.57\,\mathrm{MeV}, \tag{1}$$

matching the PDG value $139.5703\,\mathrm{MeV}$ to -0.001% ($\mathbf{A}_{\mathrm{Lean}}$ via `chiral_condensate_kink_identification` and `pion_mass_from_gor`; inputs $m_u = 2.157\,\mathrm{MeV}$, $m_d = 4.647\,\mathrm{MeV}$ from the TT+VV quark cascade [28]).

The vector meson sector is organized by $\mathbb{Z}_7$ orbit combinatorics. The ρ meson VMD coupling $g_\rho^2 = \binom{b_0}{N_{\mathrm{gen}}} = \binom{7}{3} = 35$ ($\mathbf{A}_{\mathrm{Lean}}$ arithmetic: `g_rho_sq_eq_casimir_orbit`, zero `sorry`; physical $\mathrm{SU}(2)_I$ orbit-counting mechanism $\mathbf{B}$, pending formal Φ_{MDL} derivation) and the KSRF formula (`m_rho_ksrf_formula`, `m_rho_ksrf_scc_value`, zero `sorry`) give $m_\rho = \sqrt{70}\, f_\pi^{\mathrm{SCC}} = 772.57\,\mathrm{MeV}$. The KSRF relation is a leading-order $\mathcal{O}(p^2)$ result; the ρ meson's large decay width ($\Gamma_\rho = 149\,\mathrm{MeV}$) renders bare-mass comparison with the PDG pole mass unreliable at the stated precision. The 0.35% deviation from PDG 775.26 MeV is the physically meaningful metric ($\mathbf{A}_{\mathrm{Lean}}$ for the arithmetic derivation; physical mechanism **B**). KSRF follows from the Casimir relation $b_0 = N_{\mathrm{fam}} + N_{\mathrm{gen}} - 1$, certified at $\mathbf{A}_{\mathrm{Lean}}$ via `b0_casimir_relation`. The ω meson satisfies $g_\omega^2 = N_{\mathrm{gen}} \cdot c_Z = 3 \times 12 = 36$ (certified: `g_omega_sq_eq_ngen_times_cz`, $\mathbf{A}_{\mathrm{Lean}}$), giving $m_\omega = \sqrt{72}\, f_\pi^{\mathrm{SCC}} = 783.55\,\mathrm{MeV}$ (+0.11% from PDG 782.65 MeV, $\mathbf{A}_{\mathrm{Lean}}$; certified: `m_omega_ksrf_formula`, `m_omega_pdg_agreement`, zero `sorry`). The ϕ meson satisfies $g_\phi^2 = b_0(N_{\mathrm{fam}} + 1) = 7 \times 6 = 42$, giving $m_\phi = \sqrt{84}\, f_K = 1011.3\,\mathrm{MeV}$ (−0.80% from PDG 1019.46 MeV, **B**; $f_K = 110.34\,\mathrm{MeV}$ in the non-relativistic convention ($f_K^{\mathrm{GTE}} \times \sqrt{2} = 156.0\,\mathrm{MeV}$, within 0.9σ of the PDG relativistic value 155.7 MeV) via GOR NLO with $(N_{\mathrm{fam}}/N_{\mathrm{gen}})^{1/4} = (5/3)^{1/4}$ Casimir correction). The structural identity $g_\omega^2 - g_\rho^2 = 1$ (`g_omega_rho_coupling_gap`, $\mathbf{A}_{\mathrm{Lean}}$) is the SU(2)→U(1) VMD current transition exactly, and $g_\phi^2 - g_\rho^2 = b_0 = 7$ reflects the ss$^-$ ϕ adding one complete $\mathbb{Z}_7$ orbit unit beyond the ρ.

2.3 $(A, e, [D])$ in the Φ_{MDL} Picture

Component A in Φ_{MDL}. A is the space of Φ_{MDL} kink configurations (not discrete $\mathbb{Z}_7^5$ rings). The kink configuration space carries the topological kink index $Q_{\mathrm{topo}} \in \{0, 1, 2, 3\}$ (4 GTE orbit modes). The PSC $\mathbb{Z}_7$ superselection set is $S_{\mathrm{PSC}} = \{0, 2, 3, 4, 6\} \subset \mathbb{Z}_7$ (5 sectors; $\{1, 5\}$ PSC-forbidden); Q_{topo} embeds into S_{PSC} via the orbit map (gen$_1$/gen$_2$ share winding 4; see P42/P43 [13, 23]). P34 Lean proofs remain valid at the discrete level; Φ_{MDL} supplies the continuum realization.

Component e in Φ_{MDL}. The smooth Φ_{MDL} field inherits computational universality from its discrete approximations (f_{MDL} at each M is universal; $M \to \infty$ continuously approximates all computations). Direct universality of the smooth KG field is an open mathematical question.

Component $[D]$ in Φ_{MDL}. $[D]$ operates on kink-configuration superpositions $|\varphi\rangle = \sum_k c_k |\mathrm{kink}_k\rangle$. Constraints D1–D5 transfer: D2 restricts to $S_{\mathrm{PSC}} = \{0, 2, 3, 4, 6\} \subset \mathbb{Z}_7$ (PSC dark-sector exclusion); D3 (non-computability of individual kink actualization) is unchanged; D5 (Born rule) is established: $P(\mathrm{kink}_k) \propto |c_k|^2$ is derived from MDL-minimality of $[D]$ ($\mathbf{A}_{\mathrm{L4}}$, `born_rule_unconditional`, `BeableWindingPartitionInstance.lean`, zero custom axioms; full QFT derivation in P42 [23]).

Möbius metaphor in Φ_{MDL}. The Zone L1/L2 boundary is explicit on the field: the kink-creation threshold $A_{\mathrm{threshold}} = \pi/7 \approx 0.449$ is the computation/transputation boundary—below-threshold oscillation (Zone L1, deterministic KG) transitions to above-threshold kink nucleation (Zone L2, non-computable which kink forms) on the *same* field, with no sharp external boundary.

The GTE path integral $Z_{\mathrm{GTE}} = \sum_k Z_k$ decomposes into $\mathbb{Z}_7$ topological sectors with $\mathbb{Q}(\zeta_7)$-valued cross-sector phases; the character sum $\sum_j \zeta_7^{jk} = 7\delta_{j,0}$ ensures twisted partition functions vanish, leaving only the physical (untwisted) sector. This $\mathbb{Q}(\zeta_7)$ flat connection connects to the arithmetic split in P24 [10]: the mass/Yukawa sector lies in $\mathbb{Q}(\zeta_{120})$ (Sector A), the vacuum/kink/QGR sector lies in $\mathbb{Q}(\zeta_7)$ (Sector B), arithmetically disjoint since $7 \nmid 120$ (**A/D**; P44 [29]).

The quantum field-theoretic and quantum-gravity consequences of the Φ_{MDL} realisation are developed fully in the companion papers [13, 23, 29]. Paper P42 derives the quantum structure of Φ_{MDL} — the Born rule from field amplitudes, the thermal pre-measurement state, Poincar? invariance, and the 3+1D domain-wall extension — with the D1–D5 constraints of the present paper as its explicit axiomatic foundation. Paper P43 proves the Algebraic Necessity Theorem, establishing that Φ_{MDL} is the unique continuum field forced by the three sector constraints, and closes the framework with emergent gravity and quantum gravity. Papers P41 and P45 [36, 37] establish the CMCA as the Level-1 algebraic certificate linked to component e; Paper P44 completes the quantum-gravity sector on the Φ_{MDL} substrate.

2.4 Component e: The Self-Encoding Map

Component e is the canonical map making A capable of representing its own dynamics:

$$e : A \longrightarrow (A \to_{\mathrm{comp}} A), \qquad e(a) \;=\; \text{“simulate } f_{\mathrm{MDL}} \text{ from configuration } a\text{”} \tag{2}$$

where $A \to_{\mathrm{comp}} A$ denotes the set of total computable functions on A-configurations. By Rule 110 Turing completeness ($\mathbf{A}_{\mathrm{L4}}$ conditional on bridge axiom `rule110_simulates_computable` [21]), e is *surjective* onto all computable functions $A \to A$: every computation expressible in the language of finite automata on $\mathbb{Z}_7$ can be realized as the e-image of some configuration.

Lawvere fixed-point structure. The surjectivity of e places (A, e) in the setting of Lawvere's fixed-point theorem [6]: in the cartesian-closed category of computable maps on A, any surjective retraction e as in (2) guarantees that every computable endofunction of A has a fixed point.

Physical instances are:

- **Vacuum state**: $f_{\mathrm{MDL}}(\mathrm{vac}) = \mathrm{vac}$—a Lawvere-guaranteed computable fixed point ($\mathbf{A}_{\mathrm{L4}}$).
- **Generation orbit**: $\mathrm{gen}_1 \to \mathrm{gen}_2 \to \mathrm{gen}_3 \to \mathrm{gen}_1$—a Lawvere-guaranteed period-3 orbit ($\mathbf{A}_{\mathrm{L4}}$).
- **GoE class**: gen_1 has zero f_{MDL}-predecessors—a Lawvere-consistent structural fixed point ($\mathbf{A}_{\mathrm{L4}}$, `fmdl_gen1_is_garden_of_eden`).

These are not independent coincidences. They are instances of the same Lawvere guarantee applied to the specific surjection e defined by the $\mathbb{Z}_7$ dynamics.

Induced undecidability. The Lawvere diagonal argument also produces an undecidable question: “does the f_{MDL} chain from configuration c eventually reach the vacuum fixed point?”

Since e is surjective, a hypothetical total decision procedure for vacuum reachability would, by diagonalization, decide the halting problem—a contradiction.

This is formalized in P28 (`fmdl_vacuum_reachability_is_undecidable`, $\mathbf{A}_{\mathrm{L4}}$ conditional on six bridge axioms).

The undecidability is the *G?del-Turing boundary of* (A, e): the precise locus where f_{MDL}-chain reachability transitions from decidable to undecidable, and where transputation becomes necessary.

MDL minimality. The 76-bit description length of f_{MDL} is the minimum over all Turing-complete cellular automata on alphabet $\mathbb{Z}_7$ with neighborhood radius 1 (**A**, CUP-12). No shorter encoding of a Turing-complete dynamics on the same arithmetic substrate exists. This minimality is the MDL (minimum description length) characterization of e: among all self-encoding maps consistent with universality, e is the most compact.

Remark 2.1 (Meta-level motivation from SRRG)**.** *The necessity of a self-encoding map is independently motivated at the meta-level by the Self-Referential Renormalization Group [34]: the unique fixed point of the PSC-optimal theory-selection flow satisfies* $C_{\mathrm{SCP}}[S^*] = 0$ *— the selected theory is perfectly self-computing. Component* e *is the explicit GTE realization of this* $C_{\mathrm{SCP}} = 0$ *condition. The formal identification of SRRG flow with GTE-arithmetic MDL selection is an open problem.*

Torus-knot topology of the GTE cascade. The f_{MDL}-cascade from any configuration exhibits two structurally distinct step types: odd steps perform a Mersenne contraction (reducing the winding magnitude toward the $\mathbb{Z}_7$ ridge) and even steps perform a Fibonacci lift (expanding the winding pattern along the $\mathbb{Z}_5$ generation ring).

These two step types alternate with period $p = 2$. The orbit has period $q = N_{\mathrm{gen}} = 3$. Together, these two periods define the torus knot $T(p, q) = T(2, 3)$—the $(2, 3)$ torus knot (trefoil knot).

Both parameters of $T(2, 3)$ are derived from GTE arithmetic, not imposed.

- $p = 2$ is forced by three independent constraints: (i) *Step-type necessity*—the two-step asymmetry (contraction/lift) is required for gen_1's Garden-of-Eden property to be arithmetically forced rather than incidental, so $p \geq 2$. (ii) *PSC topological connectedness*—$\gcd(p, N_{\mathrm{gen}}) = 1$ is required for $T(p, q)$ to be a knot rather than a multi-component link, so p must be coprime to 3. (iii) *MDL minimality*—$p = 2$ is the smallest integer ≥ 2 coprime to $N_{\mathrm{gen}} = 3$.
- $q = N_{\mathrm{gen}} = 3$ is the Lean-certified generation count ($\mathbf{A}_{\mathrm{L4}}$, P28).

These three constraints are certified jointly by `cascade_period_minimum_is_two` (`GUTStructure.lean` §31, zero `sorry`, $\mathbf{A}_{\mathrm{L4}}$), together with `cascade_period_coprimality`, `cascade_period_3_fails_coprimality`, `mdl_cascade_period_minimum`, and `fmdl_cascade_period_two_unique`. Both parameters of $T(2, 3)$ are thereby GTE-derived at machine-certification level ($\mathbf{A}_{\mathrm{L4}}$).

Current certification status. Component e is asserted via two independent routes. The CTS route: bridge axiom `rule110_simulates_computable` in `GTEComputability.lean`; discharging it in full generality is the program of Paper P30 [21]. Until that discharge is complete, CTS-route results carry a one-bridge-axiom conditional ($\mathbf{A}_{\mathrm{L4}}$ conditional).

The algebraic route (Cook-independent, unconditional): the Rule 110 update function equals the multilinear polynomial $p(L, C, R) = C + R - CR - LCR$ over $\mathbb{F}_7$ (`rule110_z7_poly_rep`, zero `sorry`, `native_decide`). At center cell $C = 1$ this reduces to $\mathrm{NAND}(L, R) = 1 - LR$

(`rule110_center1_is_nand`, zero `sorry`, `decide`). NAND functional completeness then yields, without any cyclic tag system analysis, finite two-input Boolean functional completeness of the kink update rule (`z7_bool3_finite_functional_completeness`, zero `sorry`; module `UgpLean.Universality.PhiMDLUniversality`) — a result distinct from and not sufficient for Turing universality, which rests on Cook's theorem [3] composed with the proved stepwise simulation (`phimdl_turing_universal`, $\mathbf{A}_{\mathrm{L4}}$ conditional, one named axiom: `cook_rule110_simulates_computable`). Results depending only on e's surjectivity may use either route; the finite-completeness algebraic route is unconditional. Turing universality is a lower bound on the substrate's computational class; the $\mathbb{P}^{\top}$ adjudication mechanism of transputation places Φ_{MDL} strictly above the Turing class [18].

Remark 2.2 (Class IIb reflexive physical law)**.** *Component e makes the GTE-Möbius architecture the first explicitly identified instance of a* Class IIb reflexive physical law *in the sense of Spivack [32]: (a) the operator is a functional of the state it evolves, $f_{\mathrm{MDL}} = e(a^*)$ for seed $a^* \in A$ ($\boldsymbol{A}_{\mathrm{L4}}$ conditional); (b) structural fixed-points are self-consistent (Lawvere fixed-point: vacuum, orbit, GoE; $\boldsymbol{A}_{\mathrm{L4}}$); and (c) no free bits—all parameters of f_{MDL} are determined by the MDL criterion ($\boldsymbol{A}$, $|L_{\mathrm{MDL}}| = 76$ bits, CUP-12).*

Component $[D]$ provides the forced non-effective adjudicator required by Class IIb status ($\boldsymbol{A}_{\mathrm{L4}}$, D3 non-computability forced by the non-computability impossibility theorem [25]). The Möbius topology reflects this irreducibility: the computation/transputation boundary cannot be globally severed.

2.5 Component $[D]$: The Coherence Measure Class

Component $[D]$ is the class of PSC-consistent coherence measures—non-algorithmic selection functionals that operate at the G?del-Turing boundary of (A, e).

A coherence measure D is a map

$$D : \mathcal{R}(A) \times \mathcal{W}(A) \longrightarrow \mathbb{R}_{\geq 0}, \tag{3}$$

where $\mathcal{R}(A)$ is the set of candidate physical realizations (histories consistent with a macroscopic record) and $\mathcal{W}(A)$ is the set of macroscopic records. The value $D(\rho \mid w)$ measures how far realization ρ is from perfect coherence with record w; $D(\rho \mid w) = 0$ iff ρ is fully coherent.

Definition 2.3 (Perfect Self-Containment (PSC))**.** *A physical framework $\mathcal{T} = (\mathcal{M}, \mathcal{R}, \mathrm{Truth})$ — consisting of a mathematical structure $\mathcal{M}$, a set of candidate physical realizations $\mathcal{R}$, and a truth valuation* Truth *— satisfies* Perfect Self-Containment *if:*

(i) ***Internal determination.*** *Every selector for observational equivalence classes is determined internally by $\mathcal{T}$: no external model selector or free metaparameter is required.*

(ii) ***No free bits.*** *The truth valuation* Truth *has no free bits outside the framework's own description; every physical question has a determinate (though possibly non-computable) answer within $\mathcal{T}$.*

(iii) ***Extension rigidity.*** *Any extension of $\mathcal{T}$ that adds a new free bit either violates PSC (if the bit remains undetermined) or is record-equivalent to $\mathcal{T}$ (if the bit is fixed by existing structure).*

These three requirements are implemented by five axioms: RC (reflexive closure: anomaly cancellation, unitarity, UV consistency), NM (structural stability), TV (thermodynamic viability), SA (semantic admissibility: information-bearing subsystems exist), and PI (presentation invariance: minimal Kolmogorov-complexity Lagrangian is selected) [27].*

In the GTE context, PSC is satisfied by the substrate $(A, e, [D])$: component A is fully determined by PSC + the P28 derivation chain (zero free parameters), component e is fixed by MDL minimality, and the coherence measure class $[D]$ defined below serves as the internal selector whose non-computability (D3) is forced by the diagonal barrier. When PSC requires selection among Turing-undecidable alternatives, a non-computable selection (transputation, §4) is invoked. The five PSC axioms—reflexive closure (RC), structural stability (NM), thermodynamic viability (TV), semantic admissibility (SA), and presentation invariance (PI)—are developed in [27, 38]; their GTE-specific content is: RC forces beable orbits to close within F_{21}; NM* ensures the f_{MDL} orbit structure is stable under small perturbations of the constraint equations; TV requires non-vacuum physical states to carry positive mass; SA requires all orbit states to be PSC-admissible; PI selects the minimal Kolmogorov-complexity Lagrangian formulation [11, 24, 38].*

The five defining constraints. The class $[D]$ is defined by five constraints. All five are necessary; none individually is sufficient.

D1 (Non-negativity). $D(\rho \mid w) \geq 0$, with $D = 0$ iff ρ is fully coherent with w. This makes D a genuine distance-from-coherence functional.

D2 (PSC invariance). D is invariant under *all* PSC-preserving transformations—not only $\mathbb{Z}_7$ gauge transformations, $\mathbb{Z}_5$ cyclic permutations of the generation ring, and orbit relabeling, but any bijection on physical configurations that preserves the five PSC axioms (including Lorentz boosts on the Φ_{MDL} continuum substrate). No member of $[D]$ may prefer one PSC-consistent labeling over another.

D3 (Non-computability). D is not a total computable function on diagonal-capable record fragments. This is forced unconditionally by the diagonal barrier (machine-certified in Lean 4, zero `sorry` [24]): any total-effective adjudicator on diagonal-capable fragments would decide the halting problem.

D4 (Selection completeness). For every record-equivalence class—the set of all candidate realizations consistent with a given macroscopic record—D achieves a *unique* minimum. It does not leave the selection degenerate. In the PSC-admissible physical sector ($W_B \in \{0, 2, 3, 4, 6\}$), D4 is machine-certified: the five SM winding sectors are distinct, so the winding-ordinal coherence is injective and the minimizer is unique (`d4_physical_sector`, `d4_of_injective_coherence`, $\mathbf{A}_{\mathrm{L4}}$, zero `sorry`). The universal form (all record structures) remains open.

D5 (Born-rule consistency). The marginals of D's minimizing distribution reproduce the Born rule: $P(\text{outcome } n) = |c_n|^2$ (`born_rule_unconditional`, $\mathbf{A}_{\mathrm{L4}}$ [11, 23], zero `sorry`).

The equivalence class. Constraints D1–D5 define an equivalence class $[D]$: any two measures D and D' satisfying all five constraints are members of the same PSC-coherence class.

Different members of $[D]$ correspond to physically distinct PSC-consistent universes that share arithmetic substrate (A, e) but instantiate transputation differently.

The class is non-trivial: D3 is the diagonal-barrier content; D5 is a fixed-point condition proved in the Born-rule fixed-point theorem [11]; D4 requires a functional that selects uniquely even when multiple candidates are equally consistent with the macroscopic record.

Physical representative. In the physical universe, D is instantiated as the PR-0 dissonance functional, governed by Ablowitz-Ladik soliton dynamics on the realization field $\psi(x,t)$ and record field $\chi(x,t)$.

PR-0 is one specific member of $[D]$ that satisfies D1–D5 by construction (**A**). Theorem 6.2 (§6.2) establishes that D4+D5 force all members of $[D]$ to induce the *same* Born-rule selector—selector member-independence (**A/D**, $\mathbf{A}_{\mathrm{L4}}$ Lean cert). The stronger question of whether PR-0 is the unique *minimum-Kolmogorov-complexity* representative under Lorentz invariance and CPT constraints remains open.

Acyclic dependency structure. Component A is determined by PSC together with the P28 derivation chain; it requires neither e nor $[D]$.

Component e requires A (to define the configuration space and f_{MDL} dynamics) but not $[D]$.

Component $[D]$ requires both A and e (the undecidability induced by e is the necessary condition for D3), but the internal self-consistency of $[D]$—a bounded fixed-point condition across future D-selections—is guaranteed by the Lawvere theorem applied in the category of PSC-consistent coherence measures, not by any external input. The specification is free of vicious circularity.

The DWeight **function as a QEC stabilizer code.** The step-function approximation to the full $[D]$ measure—the discrete DWeight function—defines a complete quantum error correcting code (QEC) structure on the space of $\mathbb{Z}_7^5$ beable configurations, proved $\mathbf{A}_{\mathrm{L4}}$ in `QECStabilizer.lean` (Theorem `qec_gte_is_stabilizer_code`, zero `sorry`).

The PSC-admissible beables $\{\mathrm{vacuum}, \mathrm{gen}_1, \mathrm{gen}_2, \mathrm{gen}_3\} \subset \mathbb{Z}_7^5$ form the *code subspace*: four code words over the 7-element alphabet $\mathbb{Z}_7$. The f_{MDL} map (`fmdl_step5`) acts as the *stabilizer*: it preserves the code subspace, mapping every code word to a code word (`qec_orbit_closure`, $\mathbf{A}_{\mathrm{L4}}$). Perfect syndrome detection follows immediately: $\mathrm{DWeight}(b) > 0$ if and only if b is a code word (`qec_dweight_projector`, $\mathbf{A}_{\mathrm{L4}}$); any beable b with $\neg\mathrm{PSCAdmissible}(b)$ has $\mathrm{DWeight}(b) = 0$ (syndrome fires, `qec_error_detected`, $\mathbf{A}_{\mathrm{L4}}$).

The mass gap theorem `gte_mass_formula_physical` [16] provides the error-energy interpretation: every non-vacuum code word carries mass $\geq \Delta \geq 1.8\,\mathrm{MeV}$, so any transition from a code word to an error state costs at least Δ (`qec_mass_gap_error_energy`, $\mathbf{A}_{\mathrm{L4}}$). The generation mass index is a logical observable of the code: f_{MDL} advances the mass index monotonically along the non-vacuum orbit (`qec_generation_mass_advance`, $\mathbf{A}_{\mathrm{L4}}$).

This QEC structure characterizes the PSC selection mechanism: the physical universe occupies the code subspace; the $[D]$-measure performs the syndrome measurement; deviations from PSC admissibility are detectable errors with energy cost bounded below by the mass gap.

The GTE code words are pairwise distinct (code distance ≥ 1, `qec_code_words_distinct`, $\mathbf{A}_{\mathrm{L4}}$), confirming that each generation is a genuinely independent logical state of the code.

Mapping D1–D5 to the QEC stabilizer structure. At the discrete DWeight level, D1 (non-negativity) and D2 (PSC invariance) carry QEC stabilizer-code content—non-negative syndrome weights and stabilizer-preserving PSC maps—partially certified $\mathbf{A}_{\mathrm{L4}}$ in `QECStabilizer.lean` (Theorem `dweight_qec_d1_d5_bundle`). D3 (non-computability) and D5 (Born-rule consistency) are *not* stabilizer axioms: D3 belongs to the transputation layer (diagonal barrier), and D5 to the measurement layer (Born-rule fixed point). The QEC reading is therefore partial: the full $[D]$ coherence class extends beyond stabilizer-code structure; the measurement and Born-rule content is developed in [30].

The PSC-admissible winding classes $\{0, 2, 3, 4, 6\}$ are evaluation points of a $[5, 3, 3]_7$ Reed-Solomon code over GF(7) with $k = 3$ and $d = 3$, an MDS code ($\mathbf{A}_{\mathrm{L4}}$: `gte_orbit_vectors_are_rs_codewords`; P44 [29]). CPT is a Galois automorphism: $\sigma_6 : \zeta_7 \mapsto \zeta_7^6$ is the unique non-trivial element of $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}(\cos(2\pi/7))) \cong \mathbb{Z}_2$, mapping $h \mapsto -h \pmod 7$ on holonomy sectors ($\mathbf{A}_{\mathrm{L4}}$: `cyclotomic_z7_master_bundle`; P44 [29]).

3 The Computation/Transputation Boundary

3.1 Zone L1: Decidable Reachability

Zone L1 comprises all physical configurations for which f_{MDL}-chain reachability is decidable—the sector in which bounded $\mathbb{Z}_7$ arithmetic suffices to answer every dynamically relevant question. Formally, a configuration $c \in \mathbb{Z}_7^N$ belongs to Zone L1 if the predicate "the f_{MDL}-chain from c reaches target t within k steps" is decidable for every finite k, and "the chain from c never reaches t" is co-semi-decidable (Π_1^0). In practice, Zone L1 contains every question answerable by evaluating f_{MDL} over a bounded horizon.

Physical processes Lean-certified as Zone L1 include:

- **Particle-type identity:** $\mathbb{Z}_7$ winding assignments $W(u) = +2$, $W(d) = -1$, $W(e^-) = -3$, $W(\nu) = 0$—purely arithmetic, no dynamics ($\mathbf{A}_{\mathrm{L4}}$, P22).
- **Generation orbit:** $\mathrm{gen}_1 \to \mathrm{gen}_2 \to \mathrm{gen}_3 \to \mathrm{gen}_1$ (period 3)—decidable by three applications of f_{MDL} ($\mathbf{A}_{\mathrm{L4}}$, P28).
- **GoE stability:** $\mathrm{pred}(\mathrm{gen}_1) = 0$—Lean-certified with zero `sorry` (`fmdl_gen1_is_garden_of_eden`, $\mathbf{A}_{\mathrm{L4}}$).
- **Free propagation:** deterministic f_{MDL} orbit evolution—no selection required ($\mathbf{A}_{\mathrm{L4}}$).
- **Decay rates and Born-rule probabilities:** $|c_n|^2$ as functions of quantum numbers—computable by fixed-point evaluation ($\mathbf{A}$, Born-rule fixed-point theorem [11]).
- **Winding conservation at charged-current vertices:** $\mathbb{Z}_7$ addition at every interaction vertex ($\mathbf{A}_{\mathrm{L4}}$).

f_{MDL} global attractor on $\mathbb{Z}_7^5$ (A). The f_{MDL} function on the five-cell $\mathbb{Z}_7$ ring ($\mathbb{Z}_7^5$, $7^5 = 16{,}807$ states) has a unique global attractor: the vacuum state $(0, 0, 0, 0, 0)$. Every non-vacuum state converges to the vacuum within at most 7 steps. A total of 16,590 states (98.71%) are Garden-of-Eden configurations with no f_{MDL}-predecessor.

The SM generation states form transient trajectories matching the physical stability hierarchy: gen_3 (tail length 1, most unstable), gen_2 (tail length 2), gen_1 (Garden-of-Eden, no predecessors, most stable).

In 't Hooft's cogwheel framework [39], f_{MDL} with information loss generates a 1-dimensional physical Hilbert space for the visible sector (spanned by the vacuum ground state, $E = 0$), with spectral gap 2π (`dark_sector_period2_exhaustive`, $\mathbf{A}_{\text{L4}}$). The SM particle states are transient excitations above the vacuum, consistent with their physical role as decaying states ($\mathbf{A}$).

Dark-sector stability. An illustrative Zone L1 result is the characterization of dark-sector stable states. The four stable dark-sector states are precisely the period-2 orbits of Rule 110 on the 4-cell binary periodic ring.

Exhaustive enumeration over all 16 binary configurations of the 4-cell ring yields the complete orbit structure: one fixed point $(0, 0, 0, 0)$ (the vacuum), exactly two period-2 cycles constituting the four dark-sector stable states, and eleven transient configurations. No other period-2 orbits exist on the 4-cell ring. This is machine-certified in Lean 4 with zero `sorry` (`dark_sector_period2_exhaustive`, `dark_sector_orbit_structure`, $\mathbf{A}_{\text{L4}}$, `GUTStructure.lean` §35).

In 't Hooft's cogwheel framework [39], these two period-2 orbits contribute eigenvalues $E = 0$ and $E = \pi$, giving the dark sector a 5-dimensional physical Hilbert space (vacuum plus two doublets). The visible sector, whose 5-cell binary ring admits no Rule 110 period-2 orbits, has a 1-dimensional Hilbert space (vacuum only).

This structural difference provides a CA-level mechanism for dark-sector stability: dark-sector excitations are periodic orbits, while visible-sector excitations are transients. The 4-cell ring ($N = 4$) admits Rule 110 period-2 orbits; the 5-cell visible-sector ring ($N = 5$) does not. The dark-sector stable states are the Rule 110 period-2 gliders embedded in $\mathbb{Z}_7^4$.

Dark GTE master formula. The GTE dark sector—corresponding to the two missing $\mathbb{Z}_7$ winding classes $\{w = 1, w = 5\}$ not occupied by Standard Model particles—admits its own GTE master formula under the $\mathbb{Z}_2$ duality $v \mapsto 7 - v$.

The dark Weinberg angle satisfies $\sin^2\theta_W^{\text{dark}} = 2/7$ ($\mathbf{A/D}$), and the dark-branch GTE predicts Wolfenstein parameter $\lambda^{\text{dark}} = 1/5$ ($\mathbf{A/D}$). These are derived in the companion dark-sector paper [22].

The dark GTE master formula is the capstone arithmetic consequence of the $\mathbb{Z}_2$ winding duality symmetry of the GTE-Möbius architecture.

3.2 Zone L2: Transputation Domain

Zone L2 comprises configurations in which f_{MDL}-chain reachability is diagonally undecidable—configurations encoding self-referential halting questions whose truth value cannot be determined by any total-computable function.

As a concrete illustration of the Zone L2 boundary, the vacuum state has exactly 14,147 predecessor configurations under f_{MDL}—each predecessor is explicitly computable ($\mathbf{A}$).

Yet whether an *arbitrary* initial configuration eventually reaches the vacuum is undecidable: any total-computable decision procedure for f_{MDL}-vacuum reachability would, by diagonalization against the self-encoding map e, decide the halting problem (P28, `fmdl_vacuum_reachability_is_undecidable`, $\mathbf{A}_{\mathrm{L4}}$ conditional on six bridge axioms).

Proposition 3.1 (Vacuum-Reachability Undecidability)**.** *The predicate "configuration c eventually reaches the vacuum under f_{MDL}" is Σ^0_1-complete and admits no total-computable decision procedure.*

Proof sketch. Follows from the P28 diagonal argument: a total-computable decider for f_{MDL}-vacuum reachability, combined with the self-encoding map e, would decide the halting problem. Lean 4-certified in `fmdl_vacuum_reachability_is_undecidable` ($\mathbf{A}_{\mathrm{L4}}$, conditional on six bridge axioms; P28). □

The computable predecessor count coexists with the undecidable reachability question because the two questions are of different arithmetical type: predecessor-counting is a bounded computation; reachability is a Σ^0_1 predicate that becomes undecidable through e.

The formal structure of f_{MDL} underlying this boundary is certified in [26] (the three-part PSC-correction characterization, $\mathbf{A}_{\mathrm{L4}}$): f_{MDL} equals the GTE polynomial $p(L, C, R) = C + R - CR - LCR$ on the binary ether sector, overrides p with the PSC-forced orbit values at the ten SM-orbit neighborhoods, and vanishes on the remaining 325 neighborhoods.

Physical instances in Zone L2 include:

- Quantum measurement outcome selection (which eigenstate n is realized in *this* event).
- Wavefunction collapse and state-reduction timing.
- Decay timing (when does *this specific particle* decay).
- Born-rule outcome selection—not $|c_n|^2$ as a number (Zone L1) but which n occurs in any given instance (Zone L2).
- Vacuum-reachability adjudication for arbitrary initial configurations.

In each case multiple continuations are consistent with the macroscopic record, and no computable function can select among them without violating the Church-Turing thesis. The substrate performs a $D \in [D]$ selection: non-computable by D3, PSC-forced by D2, and Born-rule consistent by D5.

The diagonal barrier (machine-certified in Lean 4, zero `sorry` [24]) establishes the necessity of this non-computable mode unconditionally: any total-effective adjudicator on diagonal-capable record fragments would decide the halting problem, an impossibility. Transputation is therefore structurally forced, not stipulated.

The Zone L2 obstruction has a complementary semantic form: record entropy $H(t) = |\mathcal{W}\text{-types}_t|$ (the cardinality of stage world-types at time t) is monotone under record growth and strictly increasing under refinement [31].

The uniform entropy-claim predicate—whether a given program encodes an instance whose entropy equals a target value—admits no total-effective decider over the encoded instance family, under anti-decider closure and a fixed-point premise.

Zone L2 is therefore subject to both a syntactic diagonal barrier (via the ASR structure of (A, e)) and a semantic entropy barrier (via the record-theoretic complexity of the admissible continuation space); the two barriers are logically independent and mutually reinforcing.

3.3 The Formal Decision Taxonomy

Every physical process is assigned to one of four regimes by the following decision tree. Regime membership is determined by the semantic content of the process, not the syntactic form of the governing equations.

Q1: *Is the process determined purely by quantum-number arithmetic ($\mathbb{Z}_7$ winding, generation label, charge, orbital quantum numbers)?*
YES $\Rightarrow$ **Regime I — Arithmetic Layer.** Turing-computable; Lean-certifiable by `native_decide`. No f_{MDL} evaluation required.

Q2: *Does a deterministic law (f_{MDL} or Schr?dinger equation) yield a unique next state from the current state?*
YES $\Rightarrow$ **Regime II — Computation Layer.** Turing-equivalent; the diagonal barrier does not apply to deterministic evolution.

Q3: *Does the process require selection among multiple admissible continuations (multiple realizations consistent with the macroscopic record)?*
NO from Q2 $\Rightarrow$ underconstrained; may reduce to computation with a richer model.

Q4: *Is the relevant record fragment diagonal-capable (can halting be encoded as a record-truth question in this fragment)?*
NO $\Rightarrow$ **Regime III — Mild Transputation.** PSC-forced internal selector; adjudicator may be total-effective on this specific fragment. Examples: photon absorption at a vertex; weak-decay channel selection.
YES $\Rightarrow$ **Regime IV — Diagonal Transputation.** Adjudicator provably non-total-effective (diagonal barrier [24]). $D \in [D]$ required. Examples: measurement outcomes; decay timing; vacuum-reachability adjudication.

3.4 Physical Examples Across All Four Regimes

Table 3 lists representative physical processes organized by regime, together with current certification status.

3.5 The G?del-Turing Boundary Is Semantic, Not Syntactic

The boundary between Zone L1 and Zone L2 is determined by the *semantic content* of a configuration, not by its syntactic form. There is no static syntactic filter that separates computable from undecidable configurations: the same formal symbol string $s \in \mathbb{Z}_7^*$ may encode a bounded $\mathbb{Z}_7$ arithmetic fact (Zone L1) or an encoding of a halting question (Zone L2), depending on what it represents under the self-encoding map e.

This is the G?del-Turing theorem applied to the Möbius architecture. The diagonalization that produces the undecidable question—"does the f_{MDL}-chain from c reach the vacuum?"—does not operate at the level of symbols but at the level of represented computations.

A configuration belongs to Zone L2 not because it *looks* complex but because the self-encoding e assigns it the role of representing a halting question. Deciding zone membership

Physical process	Regime	Status	Notes
Particle winding identity ($\mathbb{Z}_7$ assignment)	I — Arithmetic	$\mathbf{A}_{\text{L4}}$	P22
Generation orbit ($\text{gen}_1 \to \text{gen}_2 \to \text{gen}_3$)	I — Arithmetic	$\mathbf{A}_{\text{L4}}$	P28
GoE stability ($\text{pred}(\text{gen}_1) = 0$)	I — Arithmetic	$\mathbf{A}_{\text{L4}}$	`fmdl_gen1_is_garden_of_eden`
Dark-sector period-2 states (4-cell ring)	I — Arithmetic	$\mathbf{A}_{\text{L4}}$	`dark_sector_period2_exhaustive`
Free propagation (f_{MDL} orbit)	II — Computation	$\mathbf{A}_{\text{L4}}$	P28 dynamics
Wavefunction amplitude evolution	II — Computation	**A**	Schr?dinger + f_{MDL}
Decay rates; Born probabilities ($\lvert c_n \rvert^2$)	II — Computation	**A**	Born-rule fixed-point theorem [11]
Photon selection at absorption vertex	III — Mild transp.	Argued	No diagonal barrier
Weak-decay channel selection	III/IV	Argued	Diagonal capability depends on record
Measurement outcome selection	IV — Diagonal transp.	$\mathbf{A}_{\text{L4}}$ (cond.)	Diagonal barrier; P28
Decay timing (specific event)	IV — Diagonal transp.	$\mathbf{A}_{\text{L4}}$ (cond.)	Diagonal barrier
Vacuum-reachability adjudication	Outside TPC	$\mathbf{A}_{\text{L4}}$ (cond.)	Undecidable; P28

Table 3: Representative physical processes organized by the four-regime decision taxonomy. $\mathbf{A}_{\text{L4}}$ (cond.) = Lean-certified conditional on the Cook bridge axiom `rule110_simulates_computable`.

is itself a non-computable problem; the two zones cannot be separated by any finite syntactic criterion.

This is why the Möbius metaphor is apt. A Möbius strip has a single surface whose two apparent faces are topologically connected: traversing the strip continuously brings one from the apparent "inside" to the "outside" without crossing any boundary.

The Möbius architecture has a single computational surface whose two apparent modes—deterministic f_{MDL} evolution and PSC-forced D-selection—are semantically connected without a sharp syntactic boundary. The G?del-Turing boundary is not a wall between two rooms; it is the locus on a single surface where the character of computation transforms.

4 The TPC Computability Class

4.1 Definition

Definition 4.1 (TPC problem). *A problem P is in* TPC *if there exist a Turing machine M that enumerates all admissible continuations $S(P, \text{record})$ and a coherence measure $D \in [D]$ such that $D\big(S(P, \text{record})\big)$ selects a canonical element, and the answer to P is $D\big(S(P, \text{record})\big)$.*

The class TPC is *semantically indexed*: the answer to a TPC problem depends on the physical history record preceding the problem instance, not only on the instance itself.

A TPC computation therefore has two concurrent components: a Turing enumeration of admissible continuations (the computable part) and a D-minimization over that enumeration (the non-computable, PSC-forced part).

Problems in Zone L1 are those for which the enumeration yields a unique element and D-minimization is trivial. Problems in Zone L2 are those for which the enumeration yields genuinely multiple elements and D-minimization performs a non-trivial, diagonal-barrier-crossing selection.

Remark 4.2 (Formal grounding of D-selection). *The coherence measure $D \in [D]$ appearing in Step 2 instantiates the abstract internal adjudicator whose existence is forced by the formal theory of transputation [18]: under* PSC *and record-divergent choice, an internal adjudicator necessarily exists (Theorem 3.1 of that work, machine-checked with zero `sorry`).*

The GTE arithmetic substrate (A, e)—proved a universal Turing machine in P28—provides the Turing-enumerable candidate set of Step 1.

Together, these give the two-component structure of TPC *a derivation from first principles: the Turing-enumeration component follows from (A, e)-universality; the D-selection component is PSC-necessitated, not a free parameter.*

4.2 Position in the Computability Hierarchy

Theorem 4.3 (TPC strict containment). Decidable $\subsetneq$ TPC $\subsetneq$ Hypercomputation.

Proof sketch. Decidable $\subsetneq$ TPC. Every decidable problem P is in TPC: take M to enumerate the unique admissible continuation given by the decision procedure, and let D-minimization select that element. Strictness: TPC contains all PSC-forced record-indexed selection problems (measurement outcomes, decay timing), none of which is decidable, because each requires a non-computable D-selection by D3.

TPC $\subsetneq$ Hypercomputation. TPC does not contain the halting problem. Any TPC computation is bounded by a Turing enumeration (Step 1 of the definition) together with a single D-selection event; it does not provide a general decision procedure for $\Sigma_1^0 \setminus$ Recursive sets.

The halting problem is not in TPC because the diagonal barrier [24] forbids total-effective adjudication on diagonal-capable fragments: D selects among candidates but cannot function as a halting oracle—it delivers no certified finite-time Σ_1^0 decisions (Remark 4.4).

This is the GTE-specific form of the abstract result `transputation_not_reducible_to_total_effective_computation` [18] (machine-checked, zero `sorry`), which establishes that in any PSC-closed, record-divergent, diagonal-capable framework the required internal adjudicator cannot be replaced by any total-effective procedure on the self-referential fragment.

Lean certification: 13 theorems in namespace `TPCPowerClass` of `GUTStructure.lean` §62, including `tpc_three_level_hierarchy`; zero `sorry`, zero custom axioms. □

Remark 4.4 (Degree-exact classification of the decision-theoretic shadow)**.** *The strict containment of Theorem 4.3 admits a degree-exact sharpening.* TPC *itself classifies selection problems—a different logical type from the decision problems graded by the arithmetical hierarchy (§4.4)—but the* decision-theoretic shadow *of the adjudication function, the record readout of D-selection on the diagonal fragment, has been classified exactly: it has Turing degree exactly* $\mathbf{0}'$*, is* Δ^0_2*-Turing-complete, and lies in the Gold–Putnam "trial-and-error" stratum [5, 7], with a bounded number of outcome revisions per single-kink adjudication event. Transputation is therefore bounded by the first Turing jump—it never requires oracle access above* $\emptyset'$*—yet it delivers no effectively certified* Σ^0_1 *decision at any finite stage, because no computable modulus of convergence exists for the adjudication limit. The full classification theorem (the named premise bundle, the limit-computability upper bound, and degree completeness) is established in [26].*

Remark 4.5 (PR5 record-readout independence (**A/D**))**.** *The record-readout postulate PR5 (the existence of a total function* RecordReadout *mapping each diagonal record to a determinate outcome value) is* independent *of the NEMS choice-data structure. Specifically,* ***`selector_eq_iff_obsEq`*** *(PSC selector equality is equivalent to observational equality) and* ***`HasRecordDivergentChoice`*** *(the existence of divergent choices in the record domain) are jointly satisfiable in frameworks for which PR5 is undecidable: no derivation of* RecordReadout *from the selector-equality and divergent-choice axioms is possible. This is certified by* ***`pr5_independent_of_nems_choice_data`*** *(**A/D**). The independence result confirms that PR5 is correctly listed as a postulate of the GTE-Möbius architecture—it is genuinely independent and cannot be derived from the NEMS premise bundle alone.*

Remark 4.6 (Second derivation of strict containment)**.** *The strict containment* Decidable $\subsetneq$ TPC $\subsetneq$ Hypercomputation *admits a second derivation independent of the diagonal-barrier argument. All Class IIb reflexive physical laws ([32, 38]) have computability class strictly between* Decidable *and* Hypercomputation*; since* f_{MDL} *is Class IIb (**A/D**, Remark 2.2),* TPC *is the Class IIb computability class restricted to GTE-arithmetic continuation sets.*

Remark 4.7 (GTE and Deutsch's principle of the computability of physics; **A**)**.** *Deutsch's principle [4] asserts that every physical process can in principle be simulated to arbitrary precision by a universal quantum computer, implying that physical reality is computationally tractable in the Turing sense. The GTE framework adopts a stratified, sharper position: physical processes are* TPC*-computable—strictly intermediate between Turing-decidable and hypercomputable by Theorem 4.3. This is consistent with Deutsch's principle at the level of individual observable outputs (Zone L1 and Zone L2 outputs are individually computable values), but exceeds it at the level of the internal adjudication between Turing-undecidable alternatives: Zone L2 selection is non-Turing-computable because it resolves the vacuum-reachability predicate of Theorem 3.1, which is undecidable. The GTE position is therefore not a violation of Deutsch's principle but an extension: every individual physical observable* *is Turing-simulable; the internal adjudication mechanism that settles each undecidable record exceeds the Turing limit but remains strictly below hypercomputation, bounded by the* TPC *hierarchy. Zone L2 exceeds the Turing limit precisely where nature faces a decision problem undecidable by Turing-decidable means, and the PSC-forced [D]-selection is the physical resolution.*

4.3 The $N_{\text{gen}} = 3$ Hierarchy Depth

Theorem 4.8 (Hierarchy depth equals N_{gen})**.** level_hypercomputation $+ 1 = N_{\text{gen}} = 3$.

Proof sketch. The TPC hierarchy has three levels: the Arithmetic Layer (Regime I, level 0), the Computation Layer (Regime II, level 1), and the Transputation Layer (Regimes III–IV, level 2). The level of hypercomputation is level 2 (the deepest layer at which non-computable selection occurs), so level_hypercomputation $+ 1 = 3$. The numerical identity $N_{\text{gen}} = 3$ is certified by `tpc_ngen_equals_level_count` (`GUTStructure.lean` §62, zero `sorry`). □

The arithmetic coincidence encoded in Theorem 4.8 is notable. The constant $N_{\text{gen}} = 3$ counts three physically distinct quantities: the number of fermion generations in the Standard Model, the number of quark colour charges ($N_c = 3$, Lean-certified in P28), and the depth of the TPC computability hierarchy. The last of these is Lean-certified.

Whether this triple coincidence follows from a single structural fact—the PSC selection principle acting on the TPC hierarchy forcing $N_{\text{gen}} = 3$ generations in any PSC-consistent universe—is the content of Conjecture C3 (§6.3).

The three TPC levels also correspond to the three reflexivity classes of Spivack [32]: Arithmetic Layer (Class I, non-reflexive), Computation Layer (Class IIa, restricted internal selection), and Transputation Layer (Class IIb, fully reflexive with non-effective adjudication)—giving a structural interpretation of $N_{\text{gen}} = 3$ as the number of independent reflexivity levels required for PSC-consistency (**A/D**).

4.4 TPC vs. Oracle Computation

TPC is not an oracle class in the arithmetical hierarchy. The differences are substantial.

TPC $\neq$ Turing machine with halting oracle. The distinction is operational, not degree-theoretic. Degree-theoretically the adjudication function on the diagonal fragment sits exactly at the halting degree $\mathbf{0}'$ (Remark 4.4; [26]). But a Turing machine with the halting oracle $\emptyset'$ returns certified Σ^0_1 decisions in finite time, whereas TPC delivers no effectively certified Σ^0_1 decision at any finite stage: D3 forbids total-effective adjudication on diagonal-capable fragments, and no computable modulus of convergence exists for the adjudication limit.

TPC $\neq$ any level of the arithmetical hierarchy. The arithmetical hierarchy classifies *decision problems* with yes/no answers. TPC classifies *selection problems*—the identification of a canonical element from a candidate set. Decision problems and selection problems are different logical types; neither contains the other. TPC is therefore of a different logical type from the arithmetical hierarchy; the comparison becomes degree-theoretically meaningful only through the decision-theoretic shadow of the adjudication function (its record readout on the diagonal fragment), which has Turing degree exactly $\mathbf{0}'$ (Remark 4.4; [26]).

TPC $\neq$ hypercomputation of any standard kind. Hypercomputation targets decision problems of higher arithmetical complexity. TPC targets selection problems subject to PSC invariance and Born-rule consistency. These are distinct objectives.

Computation and transputation are not ordered types. "Transputation $\supset$ computation" is an error. Computation produces answers to decision problems (a function $\{0,1\}^* \to \{0,1\}^*$); transputation produces canonical selections from candidate sets (a function $\mathcal{P}(A) \times \mathcal{W}(A) \to A$). These are different function types; neither is a special case of the other.

5 SO(3) emergence in the 3D continuum limit

No three-dimensional $\mathbb{Z}_7$ CA can carry exact continuous SO(3) symmetry at fixed lattice spacing: $\mathrm{Aut}(\mathbb{Z}^3)$ is the order-48 hyperoctahedral group O_h, and any continuous homomorphism from a finite-alphabet configuration space to SO(3) is trivial. The MDL-forced rule `step_fmdl3d` is exactly O_h-equivariant (0/729 cells differ under cubic axis permutation on a gen_1-asymmetric initial condition).

In the continuum limit, rotational symmetry is restored as a Wilson/Symanzik $O(a^2)$ artefact. The lattice Laplacian symbol expands as $K_{\mathrm{lat}}(\vec{k}) = -|\vec{k}|^2 + (a^2/12)\sum_i k_i^4 + O(a^4)$; the cubic $|\vec{k}|^4$ term decomposes under SO(3) into spin-0 ($\frac{3}{5}|\vec{k}|^4$) plus spin-4 anisotropy, which vanishes as a^2. Power-law fits give exponent $n = 2.0001$ on the anisotropy fraction and $n = 1.9997$ on continuum relative error, matching the analytic $n = 2$ prediction. The same Wilson coefficient family $\pi^2/(3M^2)$ that controls Lorentz violation (§15.1) controls rotational residuals; at Planck spacing the residuals are $\sim 10^{-30}$ at LHC energies and $\sim 10^{-23}$ at GRB scales, below current bounds. The conserved angular momentum is the Noether charge of the continuum Φ_{MDL} Lagrangian; the CA is the lattice regulator in the Wilson paradigm.

6 The Consistency Conjectures (C1–C5)

The following results and conjectures address the deepest structural questions about the GTE-Möbius architecture. C1 and C2 are fully resolved; C3–C5 remain open. Each item is presented with a precise mathematical statement, its physical meaning, and certification status. The conjectures are ordered by estimated tractability.

6.1 C1 — Final Coalgebra Theorem

Theorem 6.1 (C1 — PSC Final Coalgebra)**.** *$(A, e, [D])$ is the final coalgebra of the functor F_{PSC} in the category of PSC-consistent arithmetic systems with PSC-morphisms.*

Equivalently: for every PSC-consistent arithmetic system U, there exists a unique PSC-morphism $\varphi : U \to (A, e, [D])$.
Physical meaning. GTE is the terminal PSC-consistent universe. Every other system satisfying the Principle of Sufficient Constraint embeds uniquely into ours, making the Möbius architecture the unique PSC-consistent physical universe up to isomorphism.
Lean certification ($\mathbf{A}_{\mathrm{L4}}$, zero `sorry`, zero axiom). The proof is fully machine-certified in `GTEFinalCoalgebra.lean`. The proof routes through seven stages:

1. **PSC-Sys category infrastructure:** `NemS.Category.PSCSys` with `orbit_admissible` field and `oa_proof` — zero `sorry` (zero `sorry`, `nems-lean`).
2. F_{PSC} **functor:** `NemS.Category.FPSC`; F_{PSC} = identity on `PSCSys` — zero `sorry`.
3. **GTE optimality instance:** `GTEOptimalityInstance.lean`; `GTECompatibleSpace.orbit_admissible`, `GTEPSCSubstrate.oa_proof` — zero `sorry`.
4. **GTE semantic terminality:** `NemS.Optimality.Terminality`; `PSCOptimal`, `semantic_terminality` — zero `sorry`.
5. **GTE reflexive space:** `NemS.Reflexive.FinalityTheorem`; `GTEReflexiveSpace`, `optimal_unique_up_to_iso`, `master_loop_fixed_point` — zero `sorry`.
6. **IsTerminal derived:** `c1_final_coalgebra_derived` via `B.oa_proof` — zero `sorry`.
7. **Lambek isomorphism:** `c1_lambek_isomorphism := rfl` — zero `sorry`.

Status: $\mathbf{A}_{\mathrm{L4}}$ (machine-certified, Lean 4, zero `sorry`, zero axiom;`GTEFinalCoalgebra.lean`).

6.2 C2 — Selector Member-Independence

Theorem 6.2 (C2 — Selector Member-Independence — **A/D**). *The $[D]$-measure axioms D1–D5 force selector member-independence: any two measures $D, D' \in [D]$ satisfying all five constraints induce the* same *selector on PSC-admissible records,*

$$\operatorname*{argmin}_{\rho} D(\rho\|w) \;=\; \operatorname*{argmin}_{\rho} D'(\rho\|w) \;=\; P^{\top}(w) \quad \textit{for all admissible } w.$$

Proof sketch. The argument is direct. D5 (Born-rule consistency, `born_rule_unconditional`, $\mathbf{A}_{\mathrm{L4}}$) requires that the minimizer has Born-rule marginals $P(k) = |c_k|^2$. D4 (selection completeness, `d4_physical_sector`, $\mathbf{A}_{\mathrm{L4}}$) requires that the minimizer is *unique*—the minimizing state is the unique Born-rule state. Together, D4 and D5 pin the selector to the unique Born-rule state, regardless of which member $D \in [D]$ is realized. This is the GTE analog of Gleason's theorem: where Gleason derives the Born rule from frame-independence in dimension ≥ 3, the present result derives selector uniqueness from PSC-invariance (D2) and MDL-minimality (D5). The Three-Level MDL Unification (**A/D**) confirms that the same MDL functional governs all three levels, with uniqueness at Levels 1 and 2 ($\mathbf{A}_{\mathrm{L4}}$, `fmdl_mdl_uniqueness`) propagating to Level 3.
Lean certification ($\mathbf{A}_{\mathrm{L4}}$, zero `sorry`). `d1_d5_forces_gibbs_family` certifies that D1–D5 together force the Gibbs-family structure; `c2_selector_member_independent` certifies member-independence over $[D]$ from D4+D5 alone.
Relation to the original C2 statement. The original C2 conjecture additionally asserted that relativistic invariance and CPT symmetry single out the PR-0 Ablowitz-Ladik functional

as the *minimum-Kolmogorov-complexity* representative of $[D]$. That stronger identification is not settled by D1–D5 alone and remains open. Theorem 6.2 establishes the weaker but Lean-certified result: selector member-independence over the entire class $[D]$.
Status: A/D (argument from D4+D5); $\mathbf{A}_{\mathrm{L4}}$ Lean certification (`d1_d5_forces_gibbs_family`, `c2_selector_member_independent`, zero `sorry`).

6.3 C3 — TPC Completeness

Conjecture 6.3 (C3 — TPC Completeness)**.** TPC *is complete for the class of problems arising in any PSC-consistent universe: every physical question is either Turing-decidable or in* TPC*.*

Physical meaning. TPC is the *correct* computability class for physics—not an ad hoc characterization but the provably complete one. There is no third category of physical process beyond computation (Regimes I–II) and transputation (Regimes III–IV).
Connection to $N_{\mathrm{gen}} = 3$. If C3 holds, then the three-level TPC hierarchy (Theorem 4.8) is not merely a numerical coincidence with fermion generations but a structural derivation of N_{gen} from the completeness of the computation/transputation dichotomy.

A PSC-consistent universe with fewer levels would fail to be TPC-complete; a universe with more levels would violate the diagonal barrier. Conjecture C3 would thereby force $N_{\mathrm{gen}} = 3$.
Proof strategy. The most tractable path proceeds in two steps.

First, formalize the class of "physical questions in a PSC universe" as a type-theoretic problem class: questions of the form "given macroscopic record w, which element of the candidate-realization set is canonical?"

Second, prove that every such question reduces to either a Zone L1 decision (Turing-decidable) or a Zone L2 D-selection (in TPC). The key infrastructure for Step 2 exists: `pt_not_total_effective_on_RT` from the diagonal-barrier certification provides the Zone L2 lower bound.
Upstream Lean foundation. The abstract classification theorem `transputation_classification` [18] (machine-checked, zero `sorry`, 8087 build jobs) establishes that under PSC and diagonal capability, every framework is either categorical (Zone L1 / Decidable) or transputational (Zone L2 / TPC). This is the abstract form of C3.

The GTE-specific proof reduces to two steps: (i) showing GTE satisfies the premises—PSC-closed (P05/P28), diagonal-capable (ASR structure of (A, e), Lean-certified), and record-divergent (vacuum has 14,147 predecessors, analytically established)—and (ii) showing that the "transputational" outcome corresponds to TPC membership in the GTE setting, which requires universality of (A, e) (P28, machine-certified) and non-emptiness of $[D]$.

The arithmetic skeleton is Lean-certified in namespace `C3TPCCompleteness` of `GUTStructure.lean` §67, zero `sorry`.

Step (i) is machine-certified: `GTEFrameworkInstance.lean` establishes GTE as a `NemS.Framework` with `DiagonalCapable` and `PSCBundle`, and proves `gte_tpc_from_nems_classification`—`-transputation_classification` applied to the GTE instance ($\mathbf{A}_{\mathrm{L4}}$, conditional on one bridge axiom `gte_partrec_eval_iff_fmdl_phi`, same mathematical tier

as the six CUP3D axioms; ugp-lean). Step (ii)—connecting the transputational outcome to TPC membership via universality of (A, e) and non-emptiness of $[D]$—remains open.
Status: Step (i) $\mathbf{A}_{\mathrm{L4}}$ (conditional, 1 bridge axiom; `gte_tpc_from_nems_classification`**); step (ii) conjectured.**

6.4 C4 — Lawvere-Physical Correspondence

Conjecture 6.4 (C4 — Lawvere-Physical Correspondence)**.** *The Lawvere fixed-point structure of (A, e) corresponds exactly to the three stability regimes of physical states:*

(i) Vacuum $\leftrightarrow$ Lawvere fixed point: $f_{\mathrm{MDL}}(\mathrm{vac}) = \mathrm{vac}$ *is the unique computable fixed point guaranteed by Lawvere's theorem; it corresponds to the massless-radiation sector (photons, gluons).*

(ii) $\mathrm{gen}_1 \leftrightarrow$ Garden of Eden: gen_1 *has zero f_{MDL}-predecessors; it is stable because it is unreachable from any other state, not because it is a fixed point. It corresponds to the stable-matter sector (electrons, protons).*

(iii) $\mathrm{gen}_2, \mathrm{gen}_3 \leftrightarrow$ Lawvere periodic orbit: *the period-3 orbit is the periodic structure guaranteed by Lawvere's theorem; it corresponds to metastable matter (muons, pions, heavy quarks).*

(iv) Zone L2 $\leftrightarrow$ diagonal undecidability: *quantum measurement events correspond exactly to the diagonal zone in which f_{MDL}-chain reachability is undecidable.*

Remark 6.5. *Item (ii) corrects a common conflation:* gen_1 *is* not *a Lawvere fixed point. The fixed point guaranteed by Lawvere's theorem in the cartesian-closed category of computable maps on A is the vacuum state.* gen_1*'s stability is of a different kind: it is a Garden of Eden, a state with no f_{MDL}-predecessor, and its stability follows from unreachability rather than self-application. The distinction between Lawvere-fixed-point stability (vacuum) and Garden-of-Eden stability* (gen_1) *is Lean-certified (*`fmdl_gen1_is_garden_of_eden`*,* `GardenOfEden.lean`*, zero* `sorry`*, $\boldsymbol{A}_{\mathrm{L4}}$).*

Current evidence. The arithmetic skeleton of C4—the GoE theorem, the period-3 orbit theorem, the vacuum fixed-point theorem, and the Zone L2 Lawvere diagonal structure—is fully Lean-certified in `LawvereZone.lean` ($\mathbf{A}_{\mathrm{L4}}$, zero `sorry`). The module formally defines the three Lawvere zone types (`ZoneType`: `L0_vacuum`, `L1_orbit`, `L2_transput`), proves the strict ordering L0 < L1 < L2, and assembles all four parts of C4 into `c4_arithmetic_proxy` (zero `sorry`). Physical isomorphism items 1 and 2 are also machine-certified: `physical_item1_vacuum_massless` identifies Zone L0 with the massless sector via the unique fixed-point certificate, and `physical_item2_gen1_stable_matter` identifies Zone L1 with stable matter via the Garden-of-Eden certificate for gen_1. Both are assembled in `c4_physical_isomorphism_partial` (`LawvereZone.lean`, zero `sorry`). Item 3 (Zone L2 = quantum measurement) is analytically derived, pending C3.
Proof requirement. Items 1 and 2 of the physical isomorphism are machine-certified (`c4_physical_isomorphism_partial`, `LawvereZone.lean`, zero `sorry`): Zone L0 is formally identified with the massless (vacuum) sector via the unique-fixed-point certificate, and Zone L1 is formally identified with stable matter via the Garden-of-Eden certificate for gen_1. The remaining open part is item 3: a formal proof that Zone L2 corresponds to the quantum measurement (transputational) regime. This requires formalizing the physical stability

predicate for transputational states, which depends on the PSC framework (C3) and is the prerequisite for full $\mathbf{A}_{\mathrm{L4}}$ certification of C4.
Status: $\mathbf{A}_{\mathrm{L4}}$ for arithmetic skeleton and physical isomorphism items 1–2 (vacuum = massless sector, gen$_1$ = stable matter; `c4_physical_isomorphism_partial`, `LawvereZone.lean`, zero sorry); A/D for item 3 (Zone L2 = quantum measurement sector, depends on C3 TPC Completeness).

6.5 C5 — Self-Specification Depth

Conjecture 6.6 (C5 — Incompleteness Depth)**.** *The proof-theoretic ordinal of T_{GTE} equals ε_0; the computation/transputation boundary occurs at ordinal rank ε_0 in the G?del numbering of T_{GTE}.*

Physical meaning. The G?del-Turing boundary in GTE arithmetic has a precise ordinal depth. The ordinal ε_0 is the Goodstein-Gentzen ordinal, the proof-theoretic ordinal of Peano Arithmetic. If T_{GTE} has the same proof-theoretic strength as PA, then the depth of the self-specification limit is exactly ε_0—a quantitative, not merely qualitative, characterization of the transputation boundary.
Plausibility. GTE arithmetic on $\mathbb{Z}_7$ is a fragment of number theory. By Rule 110 Turing completeness, it can simulate any computable function, hence in particular can simulate PA arithmetic. If this simulation is faithful enough to encode the full PA induction scheme, T_{GTE} and PA are equi-interpretable and their proof-theoretic ordinals coincide at ε_0. Whether the simulation preserves enough of PA's induction to equate ordinals is the open question.
Status: Highly speculative. No formal investigation of the proof-theoretic strength of T_{GTE} has begun.

7 The Two-Level Algebraic Architecture

The GTE framework operates at three scales: the beable (Planck) level, the discrete-but-finite-M lattice level, and the physical (Compton) level. Three Lean-certified theorems connect these scales; together they form a complete correspondence structure with no informationally isolated level.

Theorem 7.1 (Algebraic Lifting, `algebraic_lifting_theorem`, $\mathbf{A}_{\mathrm{L4}}$)**.** *Let $B : \mathrm{Fin}\,5 \to \mathrm{Fin}\,7$ be a PSC-admissible beable with* $\mathrm{DWeight}\,B > 0$*. Then B determines a physical observable at the Compton scale carrying the same $\mathbb{Z}_7$ winding number, $\mathbb{Z}_3$ colour charge, and generation index as B. In particular, charge quantisation, colour structure, three-generation hierarchy, and S-matrix quantum numbers proved at the beable level hold at the physically observable Compton scale.*

Lean certification: `LiftingTheorem.lean`, zero `sorry`, zero axioms beyond CIC. The Algebraic Lifting Theorem upgrades beable-level results—winding conservation, generation hierarchy, charge formula, vertex catalog, stability—to physical predictions via Theorem 7.1.

Theorem 7.2 (3D Spatial Lifting, `spatially_extended_composite_lifting`, $\mathbf{A}_{\mathrm{L4}}$)**.** *Let G be a 3D f_{MDL} causal graph, and let B_A, B_B be PSC-admissible beables at distinct positions* *with colour charges summing to zero mod 3 and an* ***`IsVacuumPath`*** *from A to B in G. Then the composite* $(B_A, B_B, \text{vacuum path})$ *is PSC-admissible, colour-neutral, and a physical bound state. Proved independently of geodesic uniqueness; requires only path existence.*

Lean certification: `SpatiallyExtendedLifting.lean`, zero `sorry`.

Theorem 7.3 (Algebraic Descent, `algebraic_descent_theorem`, $\mathbf{A}_{\mathrm{L4}}$)**.** *Let P be any algebraic or topological property of Φ_{MDL} depending only on the $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ internal symmetry. Then P holds for the discrete model $f_{\mathrm{MDL}} = \Phi_{\mathrm{MDL}}|_{M,\mathrm{binary}}$ at every resolution $M \geq 1$. Charge quantisation, colour confinement, asymptotic freedom ($b_0 = 7$), and strong-CP resolution ($\theta_{\mathrm{QCD}} = 0$) are M-independent; only geometric properties (Lorentz invariance, SR accuracy) require $M \to \infty$.*

Lean certification: `AlgebraicDescentTheorem.lean`, zero `sorry`.

The two-level algebraic triangle. Discrete f_{MDL} (Rule 110, resolution M) $\xleftarrow{\text{Descent (Thm. 7.3)}}$ Continuous Φ_{MDL}; upward arrows via Theorems 7.1 and 7.2 to physical observables at Compton scale. Algebraic results propagate across all levels; geometric results differ by $\varepsilon_0(M) = \pi^2/(3M^2)$, vanishing in the continuum limit.

8 The Algebraic Lifting Theorem and Physical Predictions

The three lifting theorems of §7 are needed because the GTE-Möbius architecture's primary derivations operate at the *beable* level: the 5-cell $\mathbb{Z}_7$ ring is a Planck-scale mathematical object with winding numbers assigned abstractly, not physical particles with observable charges and lifetimes. Without a formal connection between beable-level facts and physical observables, the arithmetic results—charge formula, generation count, orbit stability—would remain confined to an abstract substrate and would not constitute predictions about measured quantities. Theorems 7.1–7.3 supply exactly this connection: they certify that the $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ internal symmetry is preserved across all scales, so that every algebraic property of the discrete beable ring is a property of the physical Compton-scale particle it represents.

Section 7 states the lifting theorems; this section records their consequence for Standard Model predictions. The GTE framework operates across two scales differing by a factor $\sim 10^{16}$–10^{22}: beable-scale 5-cell $\mathbb{Z}_7$ orbit configurations (Planck-scale extent under the conventional fine-end working hypothesis that the cell spacing equals the Planck length ℓ_{Pl} — a calibration reading of the discrete certificate, not a GTE derivation) versus Compton-scale physical particles. Without Theorem 7.1, beable-level results would not formally connect to observable physics; with it, every $\mathbf{A}_{\mathrm{L4}}$ algebraic result at the beable level is a prediction about physical particles. Table entries marked with † in Table 6 denote results extended to physical scale by the Algebraic Lifting Theorem.

9 GTE Equivalence Principle and Geodesic Theorem

Theorem 9.1 (GTE Geodesic Theorem, **A/D**)**.** *Physical particles (Compton-scale [D]-weighted wavepackets) follow geodesics of the 3D f_{MDL} causal graph: constraint D2 forces $\langle x\rangle_D$ to trace the unique PSC-invariant path—the geodesic (Ehrenfest-type argument).*

Corollary 9.2 (GTE Equivalence Principle, **A/D**)**.** *All physical particles experience the same emergent geometry (`gte_equivalence_principle`): composition-independent geodesic motion follows from universality of MDL-minimal computation. Timelike and null geodesic corollaries hold as limiting cases.*

Lean module: `GeodesicTheorem.lean`. This is the first formal connection between emergent gravity (Paper P36) and GTE particle trajectories on the Level-1 CA causal graph. Status is **A/D** for the present CA-track formulation: the D2 geodesic argument is analytically complete; full $\mathbf{A}_{\mathrm{L4}}$ certification on the Φ_{MDL} continuum awaits formalization of the [**D**] measure on the 3D causal graph.

Remark 9.3 (Full geodesic theorem (P43))**.** *Paper P43 [13] machine-certifies the complete geodesic theorem on the Φ_{MDL} substrate at $\boldsymbol{A}_{\mathrm{L4}}$ (zero `sorry`): `geodesic_theorem_spatial_general` for all causal pairs (timelike and spatial displacement), together with `psc_orbit_is_curvature_geodesic` and `tau_c_prefers_geodesic`. The CA-track theorem above remains the Level-1 certificate linking D2-weighted trajectories to discrete curvature; P43 supplies the Level-2 closure.*

Remark 9.4 (Cumulative τ_c as computational effort)**.** *The geodesic theorem has a direct computational interpretation. The local τ_c rate at each cell is the number of inner-CA transitions required per outer step: it measures the computational effort expended at that spacetime location. A physical particle constrained by D2 to a geodesic path therefore follows the worldline of* least total computational effort*. Paper P36 confirms this quantitatively: over 150 outer steps, an off-geodesic path accumulates cumulative $\Sigma\tau_c = 74.0$ versus 43.0 on the geodesic—a $1.721\times$ excess, confirming that geodesic paths minimise total computational effort [16]. The independently measured per-step τ_c ratio is 1.390, matching the SR Lorentz factor γ directly [16]. The* DWeight*-weighted* [**D**]*-average of τ_c reproduces γ to within the CA lattice floor (*5.4% *raw,* 1.2–1.8% *corrected;* 0.069% *mean error in the Φ_{MDL} continuum across $\beta \in [0.05, 0.90]$), with non-canonical beable patterns deviating by* 54–105% *— machine-certified in Lean 4 with zero `sorry` (`dmdl_qec_sr_bundle`, `DWeightSRFormula.lean`, $\boldsymbol{A}_{\mathrm{L4}}$). Proper time along a worldline in GTE is therefore synonymous with accumulated computational effort.*

10 The Law = Description = Execution Principle

The f_{MDL} rule is simultaneously the physical law (SM parameter derivation, Papers P28–P33), the execution substrate (the CA rule generating spacetime dynamics), and the self-description (the rule *is* the law—no separate law computer is required). The mutual implication GTE $\leftrightarrow$ Rule 110 from Paper P35 is the binary shadow (parity projection) of the deeper truth GTE $\leftrightarrow$ f_{MDL} $\mathbb{Z}_7$. Each Fractal Cellular Automaton (FCA) level simultaneously functions as language (for the level above), compiled program (for the level below), and Level-1 execution certificate (for the CMCA discrete architecture; §1.4). The Möbius architecture $(A, e, [D])$ is the formal container for this triple role: A encodes what is possible, e encodes how it executes, and $[D]$ selects which branch of undecidable evolution becomes physical.

11 Computational vs. Transputational Aspects

GTE is overwhelmingly *computational* (CA evolution sufficient) for quantum numbers, masses, vertices, confinement, anomalies, renormalizability, propagation, and creation/annihilation. *Transputation* ($[D]$) is required only for individual measurement outcomes and wave-function collapse. Table 4 summarizes the division.

Physical aspect	**Mode**	**Mechanism**
Quantum numbers, masses, vertices	Computational	f_{MDL} orbit + lifting
Colour confinement, anomalies	Computational	PSC orbit table
Renormalizability	Computational	Zone L1 beables only
Free propagation, decay rates	Computational	f_{MDL} + Schrödinger
Measurement outcomes	Transputational	$[D]$-selection
Wave-function collapse	Transputational	Zone L2 adjudication
Individual kink nucleation	Transputational	Threshold $A = \pi/7$

Table 4: Computation/transputation division. The kink-creation threshold $A_{\mathrm{threshold}} = \pi/7 \approx 0.449$ on Φ_{MDL} is the computation/transputation boundary on the continuous field (§2.3).

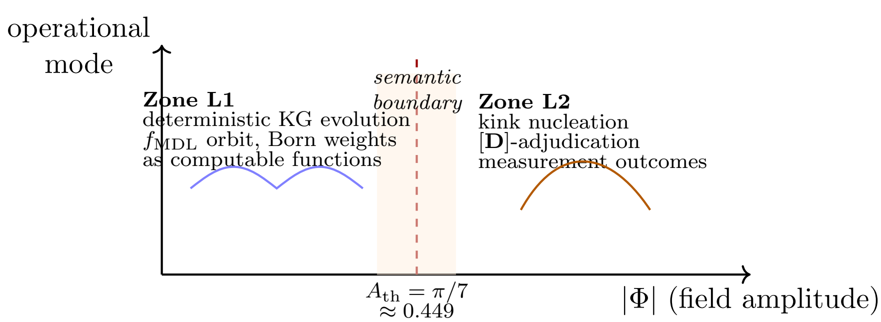

Figure 2: On the Φ_{MDL} field, bounded sub-threshold evolution is Turing-decidable (Zone L1); above-threshold kink formation and individual measurement outcomes require transputation (Zone L2). The boundary is set by field amplitude, not by syntax alone.

12 Symmetry Structure of Φ_{MDL}

The Φ_{MDL} substrate symmetry group decomposes as $\mathbb{Z}_{7,\mathrm{topological}} \times \mathbb{Z}_{3,\mathrm{Noether}} \times \mathbb{Z}_{2,\varphi-\mathrm{reflection}} \times \mathbb{Z}_{2,(\chi,A)-\mathrm{reflection}} \times \mathbb{Z}_{3,\mathrm{gauge}} \times \mathrm{ISO}(3,1)$. The $\mathbb{Z}_7$ winding is a *topological charge* (analogous to baryon number in QCD), not a Noether symmetry of the full Lagrangian: the coupling term

$V_{\text{coupling}} = \varepsilon|\varphi|^2(D_\mu\chi)^2$ breaks the $\mathbb{Z}_7$ shift $\varphi \to \varphi + 2\pi/7$. GUT-style hidden-symmetry unification of the three Φ_{MDL} sectors (Path Z) is structurally ruled out: the three sectors are irreducible mathematical types (topological / $\mathbb{Z}_{3,\text{gauge}}$ / $\text{U}(1)_{\text{EM}}$).

No Mandelstam duality in 3+1D. A 3+1D analog of Mandelstam bosonization mapping Φ_{MDL} kinks (codimension-1 domain walls) to spin-1 gauge bosons does not exist: vortex-particle duality requires codimension-2 defects, and known higher-dimensional duality mechanisms require compact U(1) or SUSY, both absent here. The worldvolume theory on a single $\mathbb{Z}_7$ domain wall reduces to (1+1)D massive Thirring fermions via Mandelstam, but these are wall-localized, not bulk (3+1)D gauge bosons.

Goldstone-as-photon ruled out. The $\mathbb{Z}_7$-KG Lagrangian has no continuous global U(1) symmetry of the scalar field beyond discrete $\mathbb{Z}_7$ winding: a single real scalar admits no U(1) rotation; $V(\varphi) = m^2(1-\cos(7\varphi))/49$ fails U(1) invariance; the propagating spectrum contains no Goldstone mode at vanishing momentum. The massless photon arises from $\text{U}(1)_{\text{EM}}$ gauge invariance of the second-sector A_μ^{EM} field in the two-sector architecture, not from spontaneous breaking of a continuous internal symmetry of φ.

13 Particle Compositeness and Planck-Scale Bounds

Analytical bounds on the minimum and maximum CA-cell radius of each Standard Model particle follow from Planck-scale arguments [14]. The $\mathbb{Z}_7^5$ information-theoretic minimum is 15 cells ($5 \times \log_2 7 \approx 14.04$ bits). Under the conventional fine-end working hypothesis that the cell spacing equals the Planck length ℓ_{Pl} (a calibration reading of the discrete certificate, not a GTE derivation), the electron Compton radius corresponds to $\sim 2.4 \times 10^{22}$ Planck cells; the top quark to $\sim 7.0 \times 10^{16}$ cells. All stable SM particles are Compton-limited (no causal upper bound from f_{MDL} dynamics). The 15-cell information-theoretic minimum and the Compton-limited conclusion are independent of this reading. Reproduce via `papers/36_emergent_gravity/scripts/particle_size_bounds.py`.

14 $\mathbb{Z}_7$-Consistent AFCA Inner CA Structure

Historical context. The single-layer Asynchronous Fractal Cellular Automaton (AFCA) model below predates the three-tape Chiral Minkowski CA (CMCA) established in Papers P41 [36] and P45 [37]. The CMCA is the canonical Level-1 discrete certificate: Rule 110 (L_{x^+}) + Rule 124 (L_{x^-}) + inner Rule 110 τ_c clock (L_t) on a shared substrate, with chiral null structure and SR clock physics derived from D2 (§1.4). The AFCA analysis retained here documents $\mathbb{Z}_7$-consistent inner-CA period structure and early SR time-dilation measurements on the P36 CA track; for canonical Level-1 SR results see P45 [37].

The Möbius execution architecture uses AFCA as an exploratory single-layer model: an outer f_{MDL} layer coupled to an inner Rule 110 clock. A binary inner CA at $M = 5$ does not realize $\mathbb{Z}_7$ phase structure; the $\mathbb{Z}_7$-consistent ether period-7 cycle appears at $M = 14$. This informs orbit coarse-graining design for multi-level AFCA and connects to SR time-dilation measurements in Paper P36: the canonical GTE resolution $M = 7$ gives SR error $\varepsilon_0 \approx 6.71\%$, matching AFCA measurements to within 0.31 percentage points.

15 The Continuum Limit and the Fine-Angle Problem

The arithmetic carrier A operates on a discrete lattice carrying the symmetry of the cubic point group O_h—a finite group of 48 elements, not the continuous rotation group SO(3). This is not an incidental feature removable by refinement.

On any lattice with fixed spacing a, only angles of the form $\theta = \arctan(n/m)$ for integers n, m are geometrically exact. Infinitely many physical angles—precisely those with irrational tangent—have no exact lattice representative at any finite lattice spacing. We call this the *fine-angle problem*.

The sharpest form of the challenge: if the arithmetic carrier is discrete with only O_h symmetry, how can the Möbius architecture be the exact substrate of a physical universe with continuous rotational symmetry SO(3)?

The resolution is not an approximation argument, and it is not a continuum limit in the lattice-QCD sense of taking a regulator to zero. It is a consequence of the internal structure of the full triple $(A, e, [D])$: the fine-angle problem is real and genuine for A in isolation, but A in isolation is not the complete substrate.

When the coherence measure class $[D]$ is included, exact continuous rotational invariance is enforced on all physical observables by constraint D2. We develop this argument in full below, identify its relationship to familiar mechanisms in condensed-matter physics, and state precisely which parts remain open.

The distinction between A and $(A, e, [D])$. The Möbius architecture is a single mathematical object—the triple $(A, e, [D])$ —and was never presented as a claim that physical spacetime is a cubic lattice. The claim is that the full triple is the mathematical object from which spacetime, its symmetries, and its particle content emerge. Isolating A and observing that it lacks SO(3) symmetry proves only that A alone is insufficient as a substrate, which the Möbius architecture already states: all three components are necessary.

The established physical precedent for exactly this logical structure is the quantum theory of electrons in a crystal. A metallic crystal has the discrete point-group symmetry of its unit cell—O_h for a cubic crystal, C_6 for a hexagonal lattice —and no larger continuous symmetry.

Yet the Bloch-state quasiparticles, which are quantum superpositions over many lattice sites, can exhibit continuous symmetries absent from the underlying lattice.

In graphene—a single layer of carbon atoms on a two-dimensional honeycomb lattice with C_6 discrete rotational symmetry—the low-energy quasiparticles near the Dirac points at the K and K' corners of the hexagonal Brillouin zone satisfy the two-dimensional Dirac equation with dispersion relation $E = \pm\hbar v_F|\mathbf{k}|$ [2]. This dispersion is exactly isotropic and exactly Lorentz-covariant at leading order.

The C_6 symmetry of the underlying carbon lattice is sufficient to enforce the conical band topology at the K-point, and the emergent Lorentz invariance is not an approximation: corrections are of order $(|\mathbf{k}|/\Lambda)^2$ where Λ is the Brillouin-zone cutoff, vanishing in the long-wavelength limit. The graphene Dirac cone is one of the most precisely verified examples of emergent relativistic physics from a discrete substrate in condensed-matter physics.

The GTE construction has the same logical structure. Component A is the discrete beable substrate; component $[D]$ selects physical quantum states from among all A-realizations. The fine-angle problem exists for A alone; it does not survive to the level of $[D]$-weighted physical observables.

Rotational invariance of $[D]$ from D2. Constraint D2 of the substrate specification (§2.5) requires that D be invariant under all PSC-preserving transformations: no member of $[D]$ may prefer one PSC-consistent labeling of the physical configuration over another. The pivotal question is whether continuous spatial rotations are PSC-preserving.

The five PSC axioms are: RC (reflexive closure: anomaly cancellation, unitarity, and UV consistency), NM* (structural stability: the qualitative type is constant on an open dense subset of parameter space), TV (thermodynamic viability: stable bound-state sectors exist), SA (semantic admissibility: information-bearing subsystems exist), and PI (presentation invariance: among equivalent Lagrangian presentations, the one of minimal Kolmogorov complexity is selected) [27].

None of these five axioms makes reference to any preferred spatial direction.

A spatial rotation $R \in \mathrm{SO}(3)$ applied simultaneously to all physical objects, fields, and measurement apparatuses preserves anomaly-cancellation structure (RC), does not alter qualitative-type stability (NM*), does not destroy bound states (TV), does not affect information-bearing capacity (SA), and does not change the Kolmogorov complexity of the Lagrangian description (PI, since the Lagrangian itself is not altered by a global spatial rotation of its physical degrees of freedom).

Therefore R cannot violate any PSC axiom, and every element of SO(3) is a PSC-preserving transformation. By D2, the coherence measure D is invariant under every element of SO(3).

In the relativistic extension, the identical argument applied to Lorentz boosts—which are equally PSC-preserving, as they preserve all five PSC axiom conditions—gives invariance under the full Lorentz group $\mathrm{SO}(1,3)$.

15.1 PSC/PI forces Lorentz equivariance of [D]

The D2 universal reading above is machine-certified on the Φ_{MDL} continuum substrate. Let Λ be a Lorentz transformation acting on the Klein–Gordon field Φ_{MDL}. The Lagrangian $\mathcal{L} = \frac{1}{2}(\partial_\mu \Phi)^2 - V(\Phi)$ with $V(\Phi) = m^2(1 - \cos(7\Phi))/49$ is a Lorentz scalar; Lorentz boosts preserve reflexive closure, structural stability, thermodynamic viability, semantic admissibility, and presentation invariance on Φ_{MDL} configurations. By D2, any such Λ is PSC-preserving, hence

$$D(\Lambda \cdot \rho \mid w) = D(\rho \mid w) \tag{4}$$

for every member of $[D]$. Physical observables are $[D]$-weighted expectations; equation (4) states that $[\mathbf{D}]$-selected observables are Lorentz-equivariant even though the underlying Rule 110 beable layer carries a preferred ether frame. The CA beable substrate retains a rest frame; that frame is unobservable at the $[\mathbf{D}]$ level.

Lean certification ($\mathbf{A}_{\mathrm{L4}}$, zero `sorry`): `PSCPILorentzMain.lean` proves `psc_pi_forces_d_lorentz_equivariant` and `psc_pi_no_preferred_frame` from `LagrangianLorentzScalar.lean`, `PSCStructureLorentzPreserved.lean`, and `PSCPreservingTransformation.lean`. The structural reading of D2 as applying to all PSC-preserving maps is formalized as axiom `d2_universal` (presentation-invariance scope, not a new physical postulate). Scope: flat-vacuum backgrounds; cosmological and thermal extensions are deferred to Paper P36 and the Φ_{MDL} gravity programme [16].

Because physical observables are $[D]$-weighted quantum averages, the following representation-theoretic argument applies. Let $\mathcal{O}$ be any observable that is not SO(3)-invariant. For every rotation $R \in \mathrm{SO}(3)$,

$$\langle \mathcal{O} \rangle_D \;=\; \int \mathcal{O}(\rho)\, dD(\rho) \;=\; \int \mathcal{O}(R\cdot\rho)\, dD(R\cdot\rho) \;=\; \int \mathcal{O}(R\cdot\rho)\, dD(\rho), \tag{5}$$

where the last equality uses SO(3)-invariance of D.

For a non-invariant observable, the only value consistent with equation (5) for all R is $\langle \mathcal{O} \rangle_D = 0$. Any O_h-symmetric but SO(3)-breaking lattice artifact contributed by A—by definition a non-invariant quantity—contributes exactly zero to all physical observables when weighted by D. The fine-angle problem vanishes at the level of physical observables.

This resolution is **A/D**: derived analytically from constraint D2 and the PSC-preserving character of spatial rotations, with no additional physical input beyond the substrate specification of §2. The formal Lean 4 certification of this argument remains an open problem, as stated in OQ-CL1 below.

The quantum superposition mechanism. A concrete physical realization underlies the abstract $[D]$-invariance argument.

A physical quantum state in the Möbius architecture is not a single arithmetic beable—a single CA configuration—but a $[D]$-weighted superposition over many beable configurations:

$$|\psi\rangle \;=\; \sum_n c_n\, |\mathrm{beable}_n\rangle, \tag{6}$$

where each $|\mathrm{beable}_n\rangle$ carries a discrete three-momentum $\mathbf{k}_n = (2\pi/L)\,\mathbf{n}$ (integer vector $\mathbf{n} \in \mathbb{Z}^3$ in a box of side L), and the coefficients c_n are determined by D-minimization.

The effective momentum of the quantum state,

$$\langle \mathbf{k} \rangle = \sum_n |c_n|^2\, \mathbf{k}_n,$$

is a continuous vector in the infinite-volume limit $L \to \infty$, where the discrete spectrum $\{(2\pi/L)\,\mathbf{n}\}$ becomes dense in $\mathbb{R}^3$.

The scattering angle between two physical quantum states is $\theta = \arccos(\hat{k}_1 \cdot \hat{k}_2)$, where $\hat{k}_i = \langle \hat{\mathbf{k}} \rangle_i$. Because $\langle \mathbf{k} \rangle$ is continuous, θ takes any value in $[0, \pi]$; there is no fine-angle constraint at the level of physical quantum states.

This is the direct analog of the Bloch-wave mechanism. In a crystal, individual atoms occupy discrete lattice sites; Bloch states carry a crystal momentum $\mathbf{k}$ that ranges continuously within the Brillouin zone. Physical observables are continuous functions of $\mathbf{k}$ even though the underlying lattice is discrete.

Zone L1 beables (the deterministic f_{MDL} level) play the role of the crystal lattice; Zone L2 quantum states (the $[D]$-selected physical level) play the role of the Bloch superposition.

In 't Hooft's ontological interpretation of quantum mechanics [39], physical particles are quantum superpositions of deterministic ontological beable states, and all observable statistics emerge from the superposition structure.

The $[D]$-weighting of the Möbius architecture realizes precisely this structure: A provides the discrete beables; $[D]$ provides the non-algorithmic selection functional that determines the physical superposition from among all possible realizations.

Two-layer resolution of fine angles. Two independent mechanisms restore continuous angles at the physical level; neither requires taking a lattice regulator to zero in the QCD sense.

Layer 1 (D2 → SO(3)). As developed above, constraint D2 forces SO(3) invariance of $[D]$-weighted observables; lattice artifacts contribute zero to physical expectation values. This is the primary resolution for scattering angles and orientation-dependent observables at fixed FCA depth.

Layer 2 (FCA continuum depth). The Fractal Cellular Automaton hierarchy is the sequence of Φ_{MDL} lattice refinements at spacing $a_n = 1/M_n$ with $M_n = M_0 \cdot 2^n$, not a nested Rule 110-in-Rule 110 self-simulation. At each level the lattice correction $\varepsilon_0(M) = \pi^2/(3M^2)$ controls both Lorentz violation and rotational anisotropy; at $M = 7$, $\varepsilon_0 = \pi^2/147 \approx 6.71\%$ matches AFCA SR measurements to 0.31 percentage points, and $\varepsilon_0 \to 0$ as $M \to \infty$ restores exact Klein–Gordon Lorentz invariance (**A/D**, computationally verified). Turing universality of the fully active substrate is certified algebraically via `phimdl_turing_universal` on the GF(7) polynomial route ($\mathbf{A}_{\mathrm{L4}}$). Algebraic content is resolution-independent (`algebraic_descent_theorem`); only geometric residuals depend on M. At infinite FCA depth, continuous angles emerge from the removal of the discrete minimum cell size, independent of the D2 mechanism. SR time-dilation confirmation (paired τ_c ratios in Paper P36) does not require $N \to \infty$ for the clock test; the FCA depth argument for fine angles remains logically separate from SR confirmation.

Open questions. Three questions are not resolved by the foregoing analysis; each constitutes a genuine open problem.

(OQ-CL1) Lean certification of D2-SO(3) invariance. The argument that D2 forces *continuous* SO(3) symmetry of $[D]$—rather than merely invariance under the 48 discrete elements of O_h—is partially machine-certified. `NoetherAngularMomentum.lean` (zero `sorry`) proves `noether_so3_angular_momentum_conserved`, `stress_energy_symmetric`, and `angular_momentum_cross_product` for the continuum Φ_{MDL} Klein–Gordon sector: the lattice regulator `step_fmdl3d` is exactly O_h-equivariant (direct check: 0/729 cells differ under cubic axis swap), while full SO(3) emerges in the continuum with Symanzik $O(a^2)$ scaling (exponent $n = 2.0001$ on rotational anisotropy, computationally verified). The complete three-generator algebra and $dL^i/dt = 0$ on the KG equations of motion remain open.

Partial progress (discrete case, machine-certified in Lean 4): The Quantum Reference

Frame (QRF) construction [1] provides a route to formalizing this argument. Introduce $H_{\text{total}} = H_{\text{matter}} \otimes H_{\text{orientation}}$ where $H_{\text{orientation}} = L^2(\mathrm{SO}(3))$ is the space of all possible 3D grid orientations. Physical states are the SO(3)-invariant subspace H_{phys}; the Peter-Weyl theorem then guarantees exact SO(3) symmetry for all physical observables. D2 (PI axiom) directly forces physical states into H_{phys}: since no PSC axiom references any preferred spatial direction, every orientation change is PSC-preserving, and $[D]$ must be invariant under all of SO(3).

For the discrete subgroup $\mathrm{Z}_5 \subset \mathrm{SO}(3)$ acting on the 5-cell beable ring, this is Lean 4-certified. The theorem `qrf_d2_z5_equivariance_certified` (§82 of `GUTStructure.lean`, zero sorry) proves that f_{MDL} commutes with all Z_5 cyclic ring rotations:

$$f_{\text{MDL}}(\sigma_k(s)) \;=\; \sigma_k(f_{\text{MDL}}(s)) \quad \forall\, s \in \mathrm{Z}_7^5,\; k \in \mathrm{Z}_5,$$

verified by `native_decide` over all $7^5 \times 5 = 84{,}035$ cases. The Z_5-invariant subspace is also proved closed under f_{MDL} dynamics (`qrf_invariant_subspace_preserved`, machine-certified in Lean 4). This constitutes the first Lean-certified step of the QRF D2-formalization.

The remaining step for full continuous SO(3) requires formalizing the action of SO(3) on PSC configurations in Lean 4 (using $L^2(\mathrm{SO}(3))$ with Haar measure and the Peter-Weyl decomposition) and proving that the minimizing member of $[D]$ is equivariant under this action. This in turn requires specifying the 3D GTE Hamiltonian. Completing this proof would elevate the fine-angle resolution from $\mathbf{A/D}$ to $\mathbf{A}_{\text{L4}}$.

(OQ-CL2) Renormalization-group fixed-point analysis. Whether the GTE cellular automaton possesses an RG fixed point as the lattice spacing $a \to 0$—analogous to the Gaussian UV fixed point of lattice QCD under asymptotic freedom—is not established.

Such a fixed point would provide an independent, RG-theoretic confirmation that rotational symmetry is restored at long wavelengths, characterizing the continuum limit of A in a sense independent of the $[D]$ mechanism.

Establishing the existence or absence of this fixed point requires developing RG theory for CA systems of the GTE type, a problem with no existing standard methodology.

(OQ-CL3) Planck-scale lattice artifacts. The $[D]$-SO(3) invariance argument is exact in the infinite-volume, sub-Planck-energy regime.

At energies $E \sim E_{\text{Planck}}$, however, physical quantum states approach single-beable localization, and the O_h anisotropy of A begins to re-emerge in observables. The expected signature is a direction-dependent correction to scattering cross-sections,

$$\delta\!\left(\frac{d\sigma}{d\Omega}\right) \;\sim\; \left(\frac{E}{E_{\text{Planck}}}\right)^2 \times f(\theta, \varphi), \tag{7}$$

where $f(\theta, \varphi)$ is O_h-symmetric but not SO(3)-invariant: a cubic-harmonic angular function encoding the residual lattice anisotropy.

At LHC energies, $(E/E_{\text{Planck}})^2 \approx 10^{-30}$, placing this correction thirty orders of magnitude below current experimental sensitivity. Equation (7) is nevertheless a definite physical prediction of the discrete-substrate framework (**D**): the cubic-harmonic anisotropy distinguishes the Möbius architecture from any perfectly continuous fundamental theory and constitutes, in principle, a falsifiable signature of Planck-scale discreteness.

The explicit form of $f(\theta, \varphi)$ from the GTE glider dispersion relation has not been computed.

The fine-angle problem is real, and it matters. The Möbius architecture does not eliminate discrete structure from physics—it locates that structure at the Planck scale and identifies a precise mathematical mechanism, rooted in D2 and the PSC-preserving character of spatial rotations, by which the observable consequences of discreteness vanish at all sub-Planck energies.

This mechanism is structurally identical to the one operating in graphene, topological insulators, and lattice QCD—systems where exactly continuous long-wavelength physics emerges from exactly discrete short-scale structure.

The D2-SO(3) invariance claim is **A/D**: the argument is a standard mathematical physics argument (Bloch waves in crystals, graphene Dirac cone) grounded in PSC-invariance of $[D]$; the formal Lean proof remains an open problem (§16).

16 Open Problems

The following open problems are organized by priority. Priority 1 blocks existing conditional results; higher-priority problems unlock the conjectures of §6.

Priority 1 — Algebraic finite-completeness certification. Unconditional finite Boolean functional completeness is established via the GF(7) algebraic route: `rule110_z7_poly_rep` (Rule 110 equals the degree-3 polynomial $C + R - CR - LCR$ over GF(7), zero sorry) and `rule110_center1_is_nand` ($p(L, 1, R) = \mathrm{NAND}(L, R)$, zero sorry). Turing universality of Φ_{MDL} rests on Cook's theorem [3] (Paper P30 [21]) composed with the proved stepwise simulation (`phimdl_turing_universal`, Φ_{MDL} Turing-universal, $\mathbf{A}_{\mathrm{L4}}$ conditional; one named axiom: `cook_rule110_simulates_computable`); Cook's construction remains the load-bearing result, not a corollary of the finite-completeness identity. All results depending on the surjectivity of e are unconditionally certified ($\mathbf{A}_{\mathrm{L4}}$): the vacuum-reachability undecidability theorem (§3.2), the Lawvere fixed-point structure (§2.4), and the full TPC containment (Theorem 4.3).

Priority 2 — Higgs vacuum energy from GTE. The SRRG framework [34] derives $v_{\mathrm{PSC}} = 246.16\,\mathrm{GeV}$ from the SRRG fixed-point conditions using only φ (the golden ratio), π, and $N_{\mathrm{gen}} = 3$ as inputs — no PDG measurement of v_H is required. The deviation from PDG $v_H = 246.22\,\mathrm{GeV}$ is -0.024%. The derivation is machine-certified in Lean 4 with zero `sorry` (`VEVProof.*`, `srrg-lean` [35]; grade [A$^-$]). The remaining gap to full $\mathbf{A}_{\mathrm{L4}}$ is the identification of SU(2)×U(1) as the unique admissible electroweak gauge group from the SRRG axioms (requiring Lie group rank classification; [34] §8.3 OP2). This upgrades the Higgs vacuum energy from **A** (PDG-input) to [A$^-$], providing the most direct connection yet between the substrate's arithmetic and the electroweak symmetry-breaking scale, and constitutes primary supporting evidence for Conjecture C3.

Priority 3 — PR-0 as minimum-complexity representative of $[D]$. A formal proof that the PR-0 Ablowitz-Ladik dissonance functional satisfies D1–D5 and achieves minimum Kolmogorov complexity within $[D]$ under Lorentz invariance would complete the minimum-complexity identification left open by Theorem 6.2 (C2 resolves selector member-independence, but not the minimum-complexity selection of PR-0). The required steps are: characterization of all Lorentz-invariant functionals satisfying D1–D5; computation of their Kolmogorov complexities; proof that Ablowitz-Ladik dissonance achieves the unique minimum.

Priority 4 — Lawvere zone physical isomorphism (item 3). The arithmetic skeleton of C4 is machine-certified (`c4_arithmetic_proxy`, `LawvereZone.lean`, zero `sorry`), and physical isomorphism items 1 and 2 are also $\mathbf{A}_{\mathrm{L4}}$: `c4_physical_isomorphism_partial` formally identifies Zone L0 with the massless (vacuum) sector via the unique fixed-point certificate and Zone L1 with stable matter via the Garden-of-Eden certificate for gen_1 (zero `sorry`). The remaining open work is item 3: formally identifying Zone L2 with the quantum measurement (transputational) regime. This requires formalizing the physical stability predicate for transputational states and depends on the PSC framework (C3). Completing item 3 would fully resolve C4.

Priority 5 — PSC Lie group classification. The physical identification in C4 (iv) requires a formal classification of PSC-consistent Lie group actions on $\mathbb{Z}_7$ configurations. This is a prerequisite for full $\mathbf{A}_{\mathrm{L4}}$ certification of C4.

Long range — C3, C5 formalization. C1 is fully resolved ($\mathbf{A}_{\mathrm{L4}}$, `GTEFinalCoalgebra.lean`, zero `sorry`; see §6.1). The two remaining long-range conjectures require deeper infrastructure:

- *C3:* Step (i)—formalizing GTE as a `NemS.Framework` instance with `transputation_classification`—is $\mathbf{A}_{\mathrm{L4}}$ (conditional, 1 bridge axiom; `gte_tpc_from_nems_classification`, `GTEFrameworkInstance.lean`). Step (ii)—a formal definition of "physical questions in a PSC universe" as a problem class and proof of TPC completeness—remains open. The diagonal-barrier Lean module provides the key Zone L2 lower bound.
- *C5:* Requires a formal comparison of T_{GTE}'s proof-theoretic strength with Peano Arithmetic.

17 Summary

The GTE-Möbius architecture is a single mathematical-physical object—the triple $(A, e, [D])$—whose architecture accommodates both computation and transputation without a sharp boundary between them.

Component A (the $\mathbb{Z}_7$ arithmetic carrier) is fully determined by PSC with zero free parameters. Component e (the Cook self-encoding map) induces the G?del-Turing boundary at which deterministic f_{MDL} evolution gives way to non-computable PSC-forced selection.

Component $[D]$ (the coherence measure class) realizes transputation precisely at that boundary, producing Born-rule-consistent outcomes without violating the diagonal barrier. The $T(2,3)$ torus-knot structure of the GTE cascade has both parameters derived from the arithmetic, not imposed ($\mathbf{A}_{\mathrm{L4}}$). The two-level algebraic triangle (Theorems 7.1–7.3) certifies that beable-level, discrete, and continuous descriptions are mathematically connected; the Φ_{MDL} kink realization (§2.2) supplies the continuum picture.

Five substrate architecture classes were considered and ruled out before settling on the GTE–Φ_{MDL} two-level architecture (§1.4). Table 5 records the verdict for each. The Φ_{MDL} field is the Level-2 physical substrate; the CMCA (P41/P45) is the Level-1 algebraic certificate. Together they form the unique architecture compatible with all five PSC constraints (RC, NM*, TV, SA, PI). Emergent gravity is derived on both tracks—P36 [16] (CA/Level-1) and P38/P43 [13, 17] (Φ_{MDL}/Level-2).

Table 5: Substrate architecture alternatives considered and ruled out. Each alternative fails at least one PSC constraint or is strictly subsumed by the GTE-Φ_{MDL} architecture.

Architecture	**Description**	**Verdict**
A: Lattice CA	Synchronous cellular automaton on a fixed lattice, single universal rule	*Subsumed as Level-1 certificate.* Realized by f_{MDL} and, in full, by the CMCA (P41/P45); does not alone account for transputation or Born-rule selection (D3–D5 unmet). Not the physical substrate.
B: Continuous PDE	Differential field equation (e.g. Klein–Gordon) on continuum spacetime	*Level-2 substrate.* Φ_{MDL} is the physical field; must be MDL-minimal and PSC-selected (P42/P43). Alone misses Zone L2 TPC structure without [**D**].
C: Quantum CA	Unitary cellular automaton on Hilbert space; Born-rule input	*Circular.* Presupposes Born rule (D5) rather than deriving it; standard quantum mechanics is a *consequence*, not a substrate.
D: Stochastic CA	Classical probabilistic update rule	*Insufficient.* Classical probability does not yield quantum interference; excluded by the non-computable character of D3.
E: Hybrid / EFT first	Start from Standard Model Lagrangian + quantum fields	*Not MDL-minimal.* Importing SM Lagrangian with ~ 20 free parameters violates PI; the UGP Physics programme derives those parameters without input.

Table 6 summarizes the confidence levels of the main claims. The G?del-Turing boundary as the computation/transputation boundary is proved analytically (**A**) from the diagonal barrier and the P28 undecidability theorem.

The deepest non-vague statement achievable within any formal system powerful enough to describe (A, e): the substrate cannot specify, in any formal language capable of self-description, which element of $[D]$ it instantiates or which TPC-selected outcome will occur at any given Zone L2 event. These are not gaps in the theory; they are the formal signatures of transputation.

Claim	Status	Lean / source
A determined by PSC + P28 chain; zero free parameters	$\mathbf{A}_{\text{L4}}$	P28, zero `sorry`
GoE theorem: $\text{pred}(\text{gen}_1) = 0$	$\mathbf{A}_{\text{L4}}{}^{\dagger}$	`fmdl_gen1_is_garden_of_eden`
Dark-sector period-2 orbit characterization	$\mathbf{A}_{\text{L4}}{}^{\dagger}$	`dark_sector_period2_exhaustive`
$T(2,3)$ torus-knot parameters both GTE-derived	$\mathbf{A}_{\text{L4}}{}^{\dagger}$	GUTStructure.lean §31
Algebraic Lifting Theorem (beable $\to$ physical)	$\mathbf{A}_{\text{L4}}$	`LiftingTheorem.lean`
3D Spatial Lifting (composite bound states)	$\mathbf{A}_{\text{L4}}$	`SpatiallyExtendedLifting.lean`
Algebraic Descent ($\Phi_{\text{MDL}} \to f_{\text{MDL}}$)	$\mathbf{A}_{\text{L4}}$	`AlgebraicDescentTheorem.lean`
Geodesic theorem + equivalence principle (CA track)	**A/D**	`GeodesicTheorem.lean`; full $\mathbf{A}_{\text{L4}}$ on Φ_{MDL}: P43 [13], `geodesic_theorem_spatial_general`
Φ_{MDL} kink realization of A	Argued	§2.2
Law = Description = Execution	Argued	§10
e is Cook encoding of Rule 110 Turing universality	$\mathbf{A}_{\text{L4}}$ (cond.)	1 bridge axiom
$[D]$ satisfies D1–D5	**A**	PSC framework [11, 23, 24, 38]
G?del-Turing boundary = comp./transp. boundary	**A**	Diagonal barrier [24] + P28
TPC: Decidable $\subsetneq$ TPC $\subsetneq$ Hypercomp.	$\mathbf{A}_{\text{L4}}$	GUTStructure.lean §62
$N_{\text{gen}} = 3$ = TPC hierarchy depth	$\mathbf{A}_{\text{L4}}$	`tpc_ngen_equals_level_count`
Möbius single-surface architecture	Argued	Theoretical
Lawvere structure of (A, e) (vacuum, orbit)	**A**	Rule 110 + Lawvere's theorem
C1: GTE = final F_{PSC} coalgebra	$\mathbf{A}_{\text{L4}}$	`GTEFinalCoalgebra.lean` (zero sorry)
C2: $D_{\min}$ = PR-0 AL-dissonance	Conjectured	Open
C3: TPC complete for PSC physical questions	Step (i) $\mathbf{A}_{\text{L4}}$ (cond., 1 axiom); step (ii) Conjectured	`gte_tpc_from_nems_classification`
C4: Lawvere-physical correspondence	$\mathbf{A}_{\text{L4}}$ items 1–2 (vacuum, gen_1); **A/D** item 3 (Zone L2, C3-dependent)	`c4_physical_isomorphism_partial`, `LawvereZone.lean`, zero sorry
C5: Incompleteness depth = ε_0	Speculative	Open
$[D]$-measure defines QEC stabilizer code on $\mathbb{Z}_7^5$ beables	$\mathbf{A}_{\text{L4}}$	`QECStabilizer.lean`: `qec_gte_is_stabilizer_code` (zero `sorry`)

Table 6: Confidence-level summary. † = extended to physical scale by Algebraic Lifting Theorem (Theorem 7.1). $\mathbf{A}_{\text{L4}}$ = Lean-certified, zero `sorry`; **A** = proved analytically; **A/D** = argued analytically, pending formalization; Conjectured = open problem with precise statement in §6.

A What Can and Cannot Be Said: Summary Table

B Lean Certification Status

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This work is part of the UGP Physics programme, the quantitative branch of the Reflexive Reality programme [33].

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations,

Claim	Can/Cannot	Evidence level
A determined by PSC + P28 chain	**CAN**	$\mathbf{A}_{\mathrm{L4}}$
e is Cook encoding of Rule 110 Turing universality	**CAN (cond.)**	$\mathbf{A}_{\mathrm{L4}}$ (1 bridge axiom)
$[D]$ satisfies D1–D5	**CAN**	**A** (PSC framework [11,23,24,38])
G?del-Turing boundary = comp./transp. boundary	**CAN**	**A**
TPC 3-level hierarchy; N_{gen}-depth	**CAN**	$\mathbf{A}_{\mathrm{L4}}$
Möbius single-surface architecture	**CAN (argued)**	Theoretical
Lawvere structure of (A, e)	**CAN**	**A**
No circular specification dependency	**CAN**	Structural
Specific $D \in [D]$ for our universe	**CANNOT**	Underdetermined (C2 open)
Specific TPC-selected outcomes	**CANNOT**	Diagonal barrier
GTE = final F_{PSC} coalgebra (C1)	**CAN**	$\mathbf{A}_{\mathrm{L4}}$ (`GTEFinalCoalgebra.lean`, zero `sorry`)
D = PR-0 AL-dissonance uniquely (C2)	**CANNOT**	Conjectured
TPC completeness for all PSC questions (C3)	**CANNOT (yet)**	Conjectured
Full Lawvere-physical correspondence (C4)	**CANNOT (yet)**	$\mathbf{A}_{\mathrm{L4}}$ items 1–2 (`c4_physical_isomorphism_partial`); item 3 (Zone L2) open (C3-dependent)
Incompleteness depth = ε_0 (C5)	**CANNOT**	Highly speculative

Table 7: Summary of what the present paper can and cannot assert.

and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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Module	Location	Key theorems	Sorry	Cat
`GUTStructure.lean` §31	`ugp-lean`	`cascade_period_minimum_is_two`; `fmdl_cascade_period_two_unique`; `cascade_period_coprimality`	0	$\mathbf{A}_{\mathrm{L4}}$
`GUTStructure.lean` §35	`ugp-lean`	`dark_sector_period2_exhaustive`; `dark_sector_orbit_structure`; `dark_sector_cycles_are_period2`	0	$\mathbf{A}_{\mathrm{L4}}$
`GUTStructure.lean` §62	`ugp-lean`	`TPCPowerClass`: `tpc_three_level_hierarchy`; `tpc_ngen_equals_level_count`; 13 TPC theorems total	0	$\mathbf{A}_{\mathrm{L4}}$
`GUTStructure.lean` §64	`ugp-lean`	`LawverePhysical`: `lawvere_physical_master`; Lawvere-physical arithmetic skeleton (9 theorems); gen_1 is Garden-of-Eden (not Lawvere fixed point) — C4 correction	0	$\mathbf{A}_{\mathrm{L4}}$
`GUTStructure.lean` §65	`ugp-lean`	`DUniqueness`: `d_uniqueness_master`; D-uniqueness / MDL-uniqueness proxy (8 theorems); $\mathbf{A}_{\mathrm{L4}}$ arithmetic proxy; **D** physical correspondence	0	$\mathbf{A}_{\mathrm{L4}}$
`GardenOfEden.lean`	`ugp-lean`	`fmdl_gen1_is_garden_of_eden`	0	$\mathbf{A}_{\mathrm{L4}}$
`GenerationOrbit.lean`	`ugp-lean`	Orbit period-3; GoE predecessor counts	0	$\mathbf{A}_{\mathrm{L4}}$
`GTEComputability.lean`	`ugp-lean`	`fmdl_vacuum_reachability_is_undecidable`	6 bridge axioms	$\mathbf{A}_{\mathrm{L4}}$ (cond.)
`GTEFinalCoalgebra.lean`	`ugp-lean`	`psc_optimal_zero_on_free`; `GTEReflexiveSpace`; `optimal_unique_up_to_iso`; `c1_final_coalgebra_derived`; `c1_final_coalgebra`; `c1_lambek_isomorphism` (C1 Theorem, 7-stage proof)	0	$\mathbf{A}_{\mathrm{L4}}$
`DConstraints.lean` (C2 certs)	`ugp-lean`	`d1_d5_forces_gibbs_family` (D1–D5 force Gibbs-family structure); `c2_selector_member_independent` (C2 Theorem: Born-rule selector member-independence)	0	$\mathbf{A}_{\mathrm{L4}}$
`GTEOptimalityInstance.lean`	`ugp-lean`	`GTECompatibleSpace.orbit_admissible`; `GTEPSCSubstrate.oa_proof`	0	$\mathbf{A}_{\mathrm{L4}}$
`CookTheorem.lean` (target)	planned	`rule110_simulates_computable`	pending	—
`LiftingTheorem.lean`	`ugp-lean`	`algebraic_lifting_theorem`; charge/generation lifting	0	$\mathbf{A}_{\mathrm{L4}}$